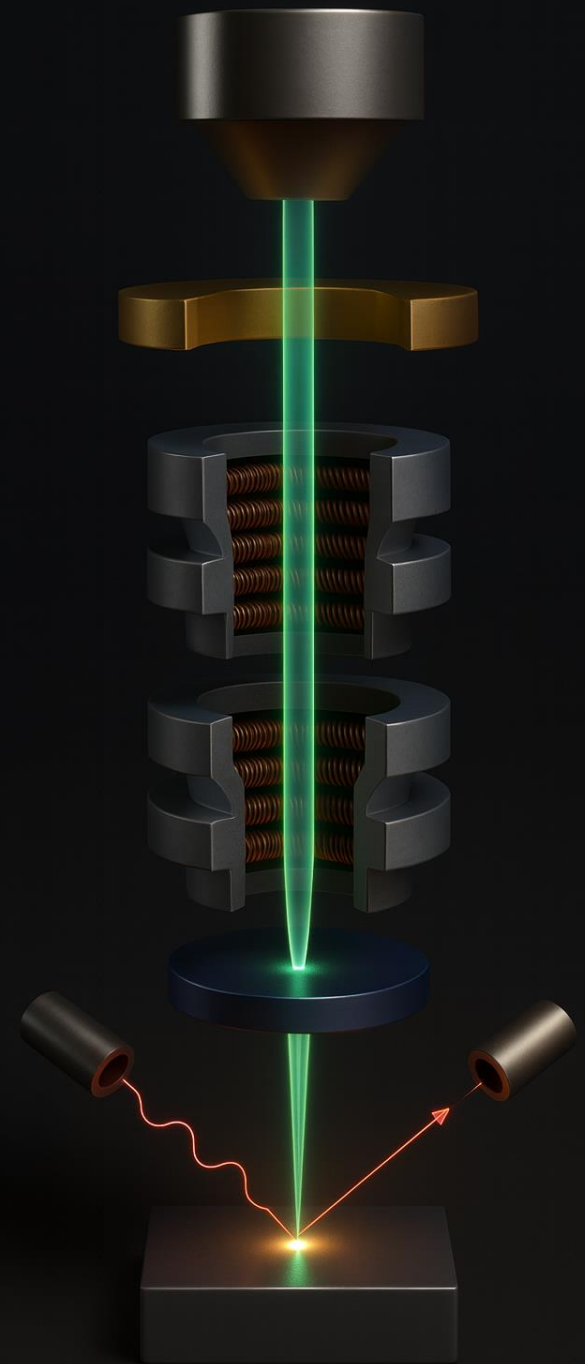
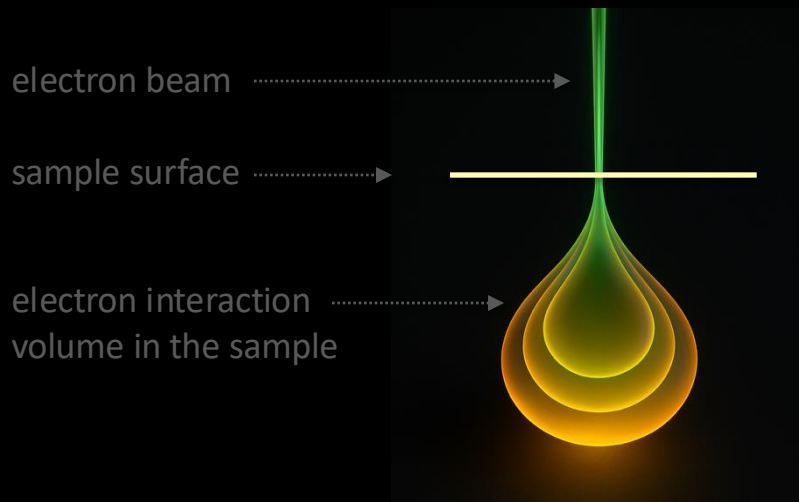
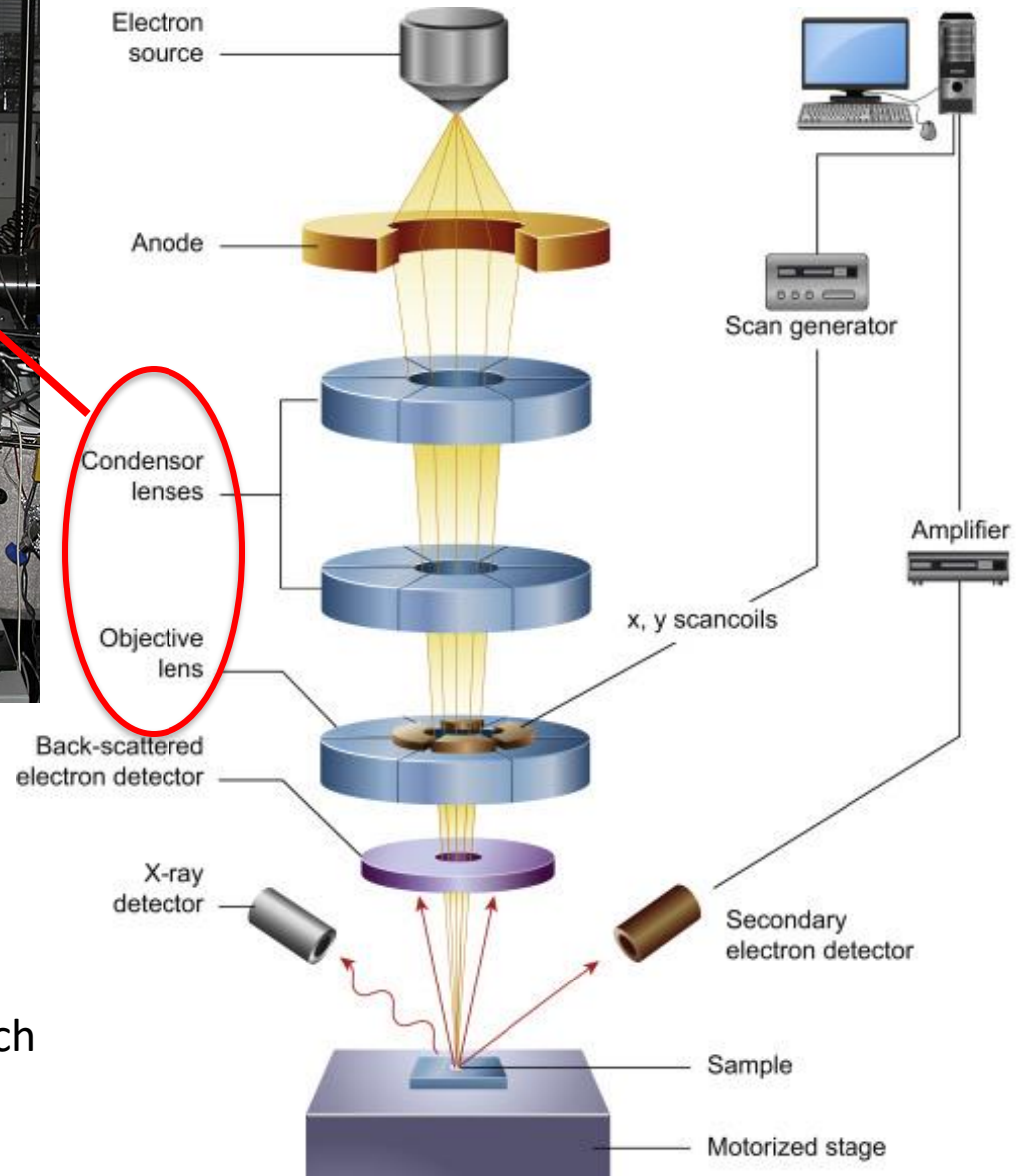
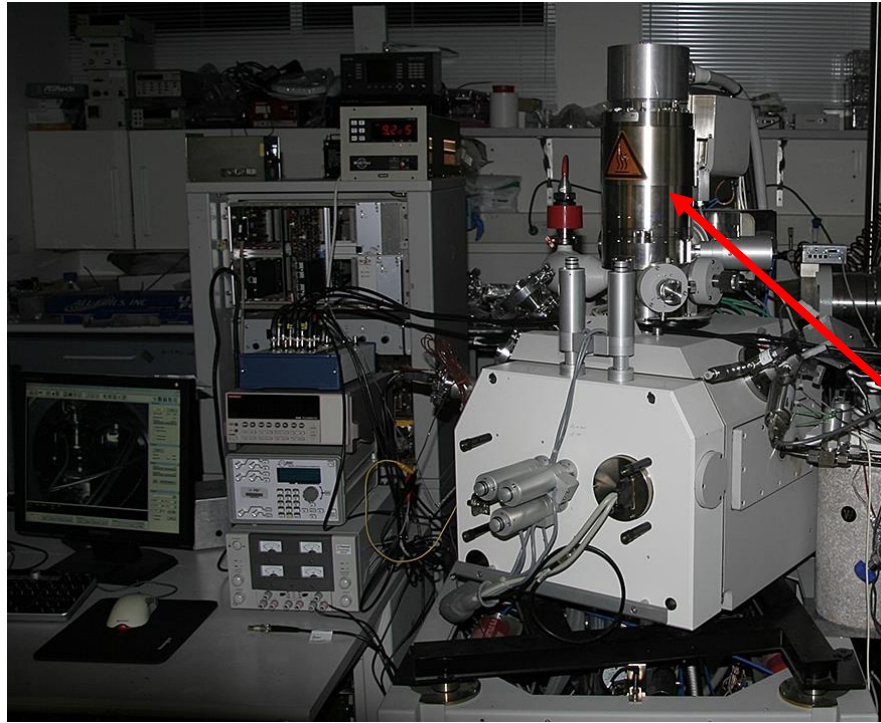


68320 Nanofabrication and Nanocharacterization Techniques

Lecture 2: Electron Optics & Electron-Solid Interactions in a Scanning Electron Microscope

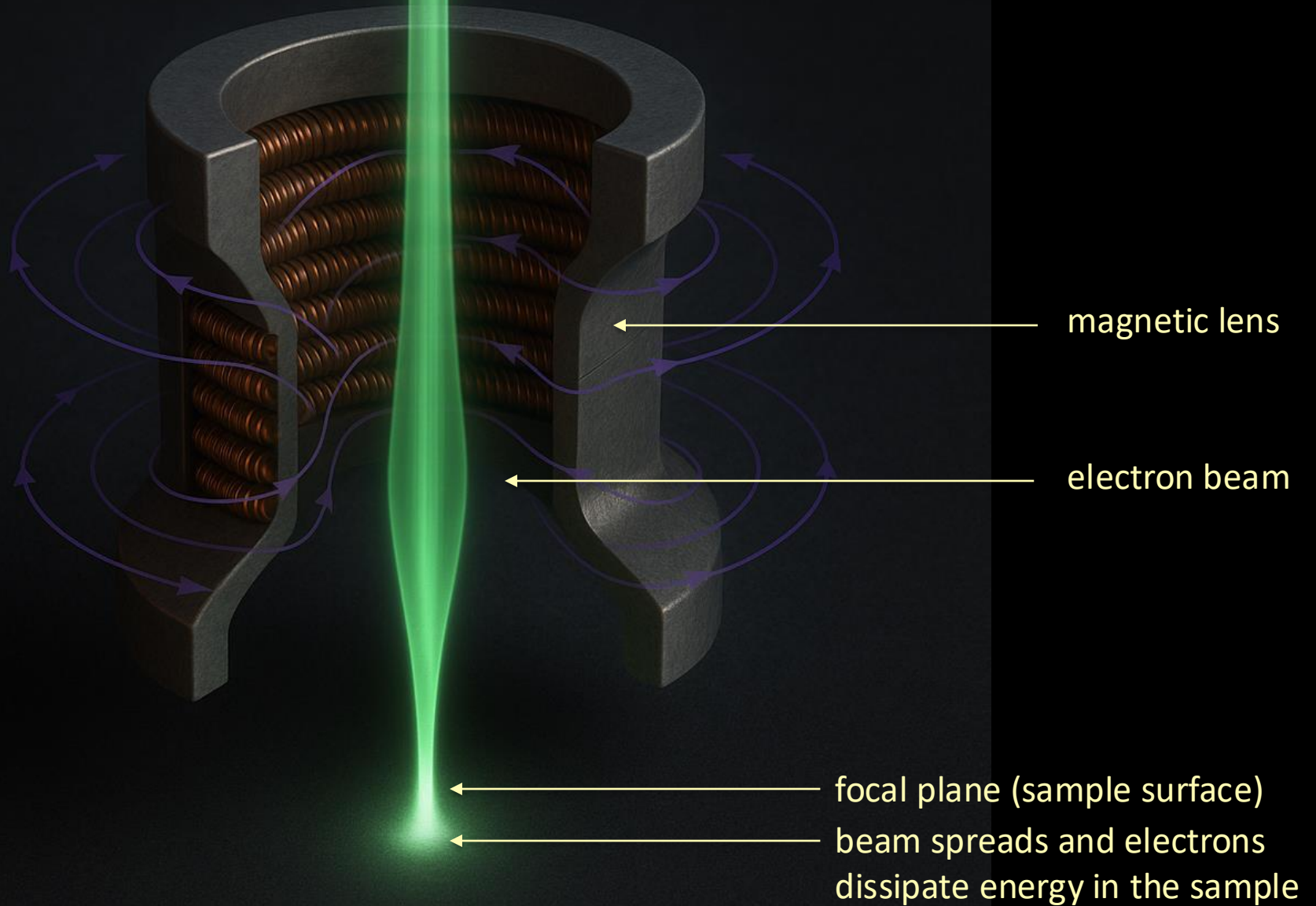


Electromagnetic lenses

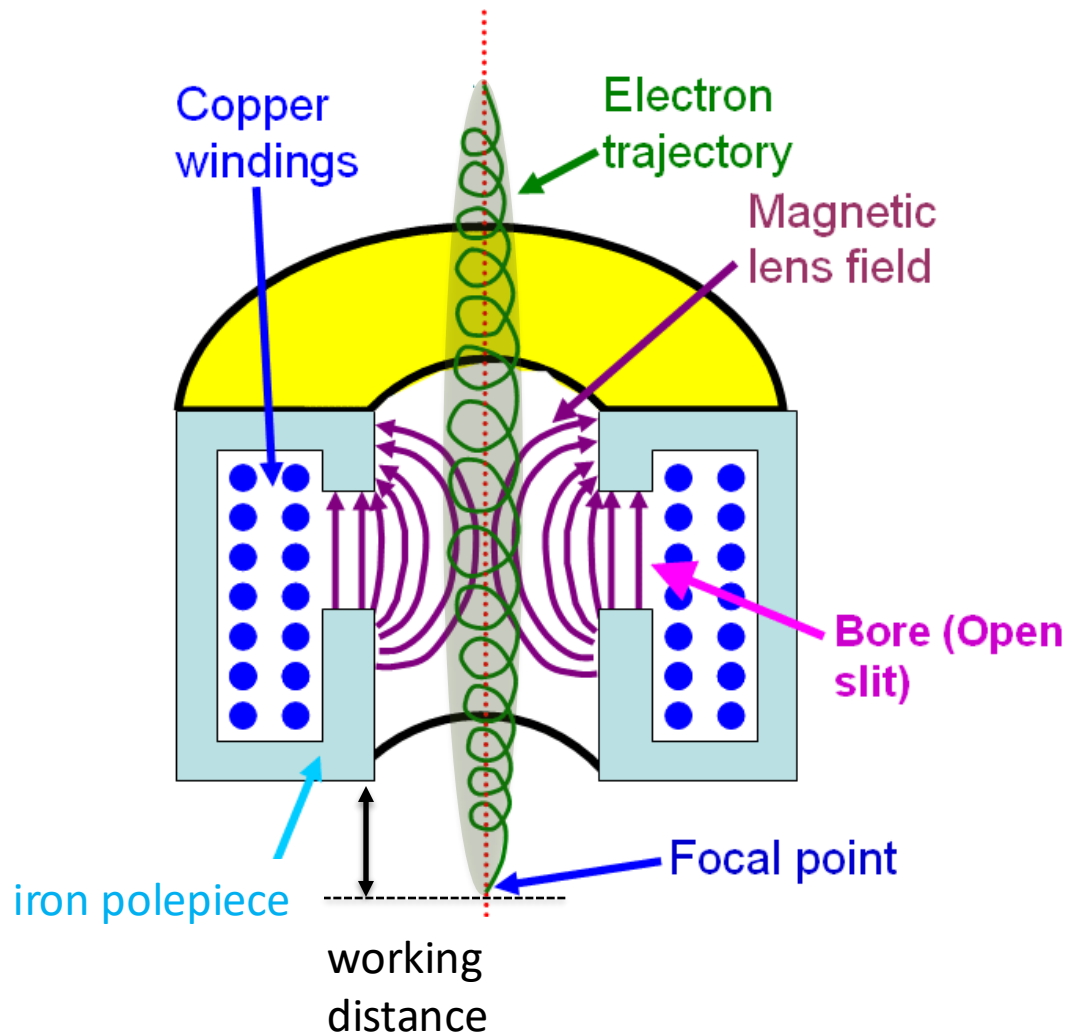


NOTE: For clarity/simplicity, this column schematic does not show apertures (which will be discussed later)

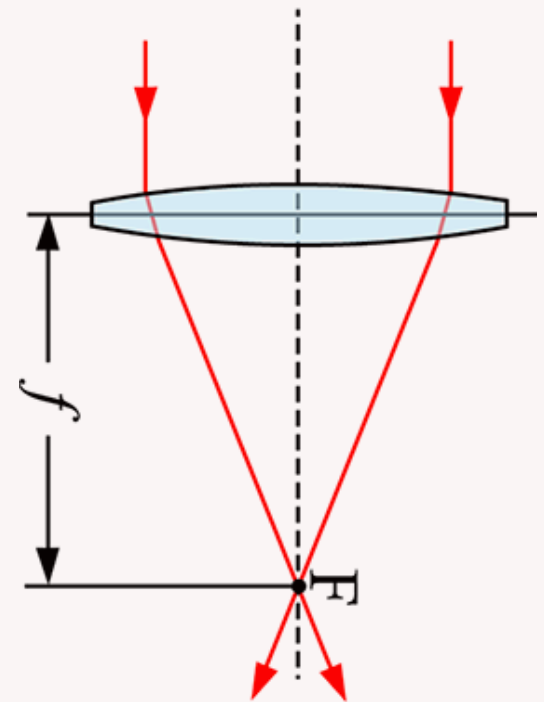
Electron lens



Electromagnetic lens



REVISION: light optics

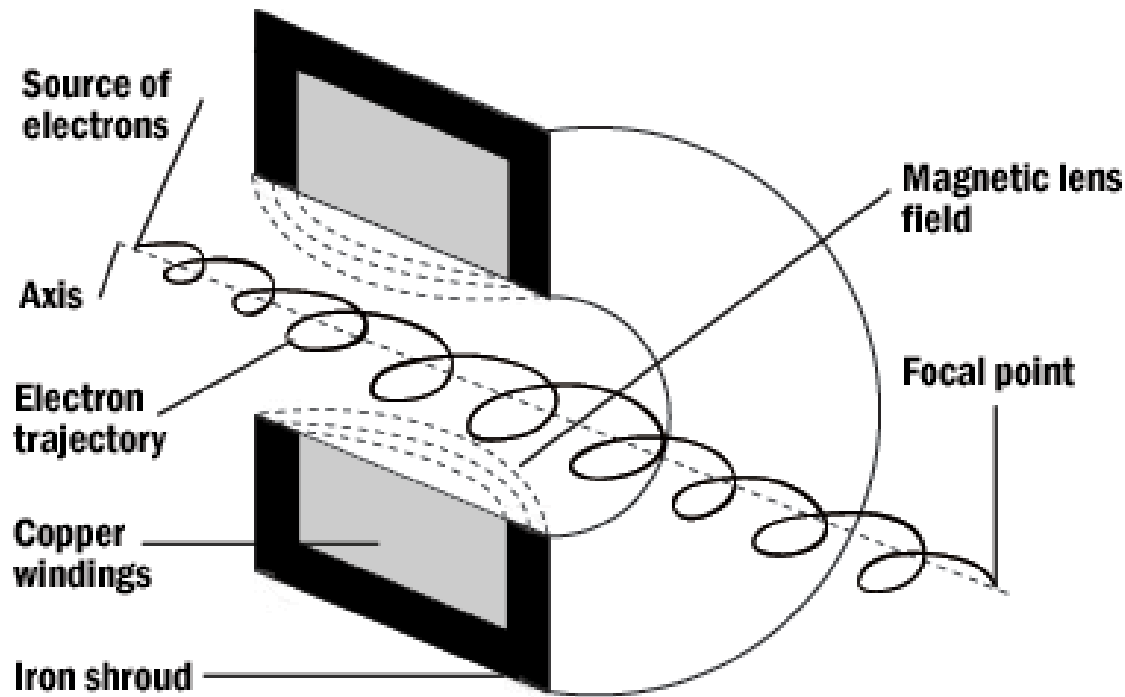


lens: glass, refractive index n
focal length, f : fixed (sample is moved up/down to focus)

$$\text{focal length, } f = K (V/I^2)$$

K = constant, V = accelerating voltage, I = current through the Cu coil (used to control focus)

Electromagnetic lens: Electron trajectories



Force exerted on an electron:

$$\mathbf{F} = q \mathbf{v} \times \mathbf{B}$$

$\mathbf{F}$ = force on a charged particle (electron)

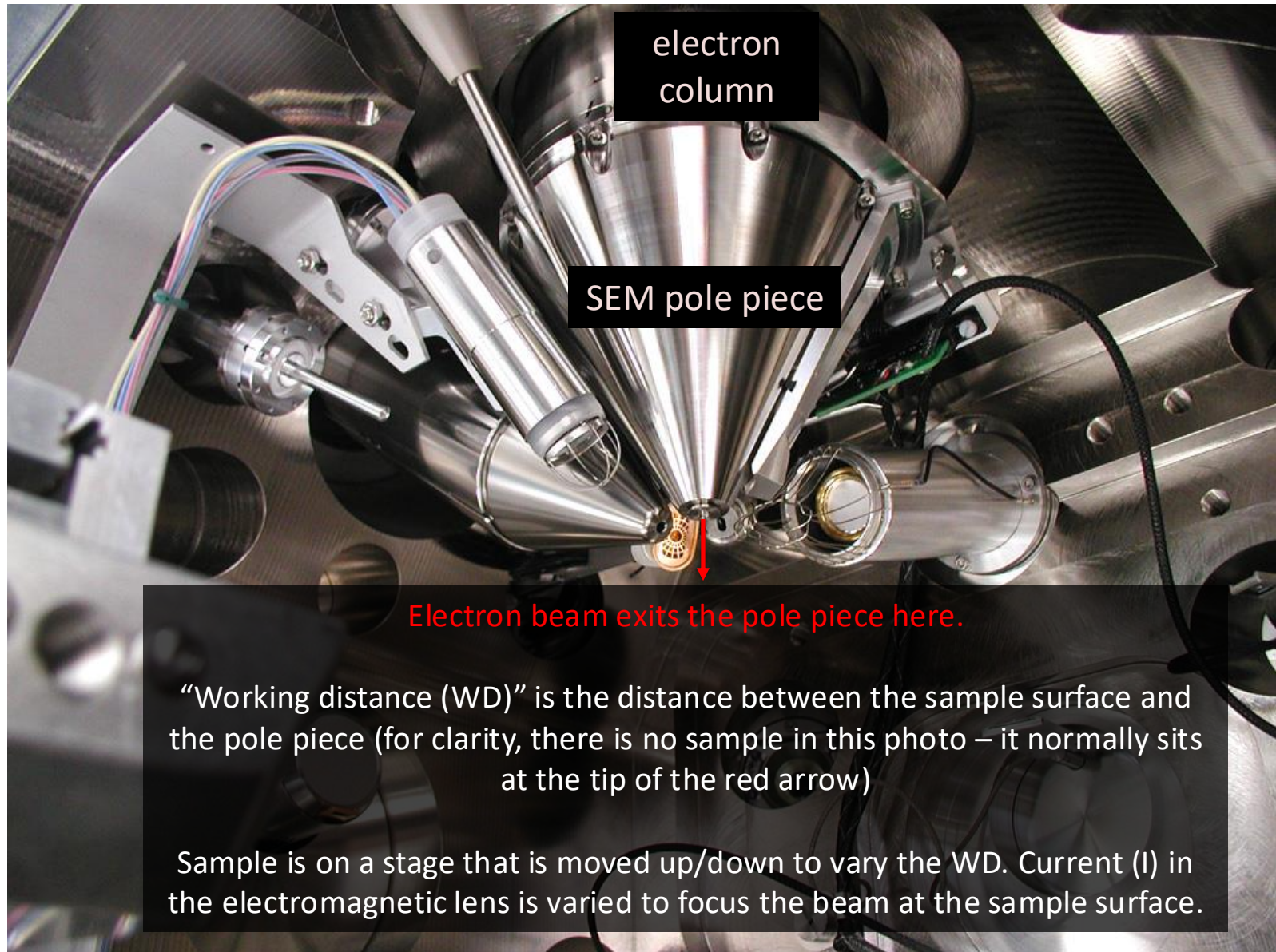
q = charge (of an electron)

$\mathbf{v}$ = velocity (of electron)

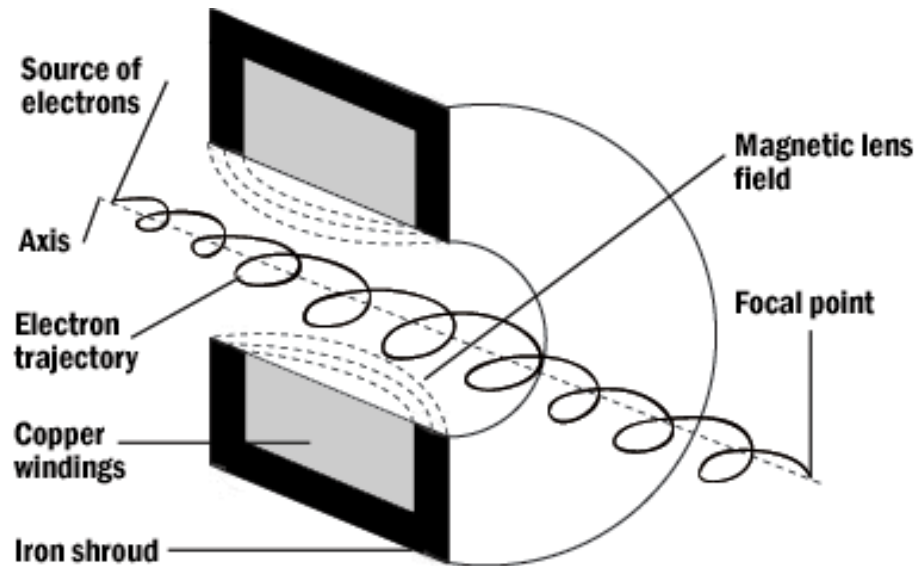
$\mathbf{B}$ = magnetic field

Working distance

Inside view of an SEM sample chamber



Sample class test question



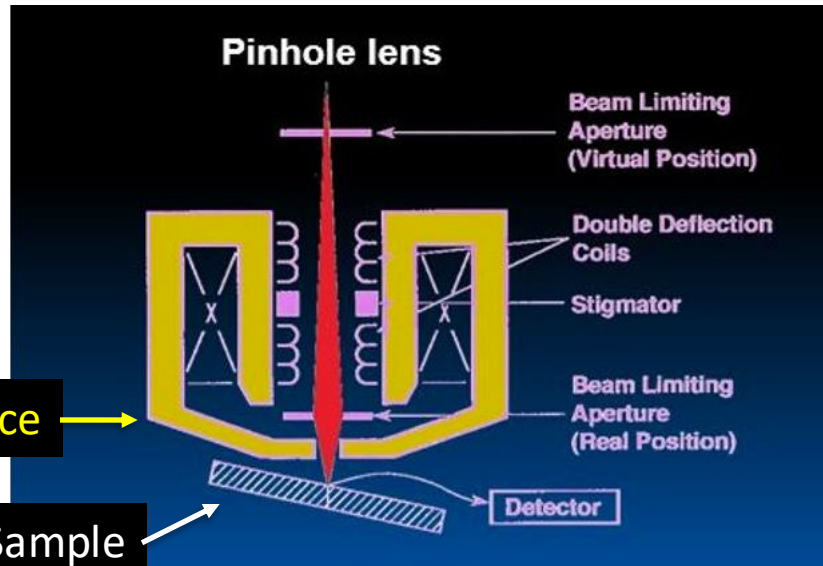
Force exerted on an electron:

$$\mathbf{F} = q \mathbf{v} \times \mathbf{B}$$

The image shows the trajectory of an electron passing through an electromagnetic lens. **Which statement is correct?**

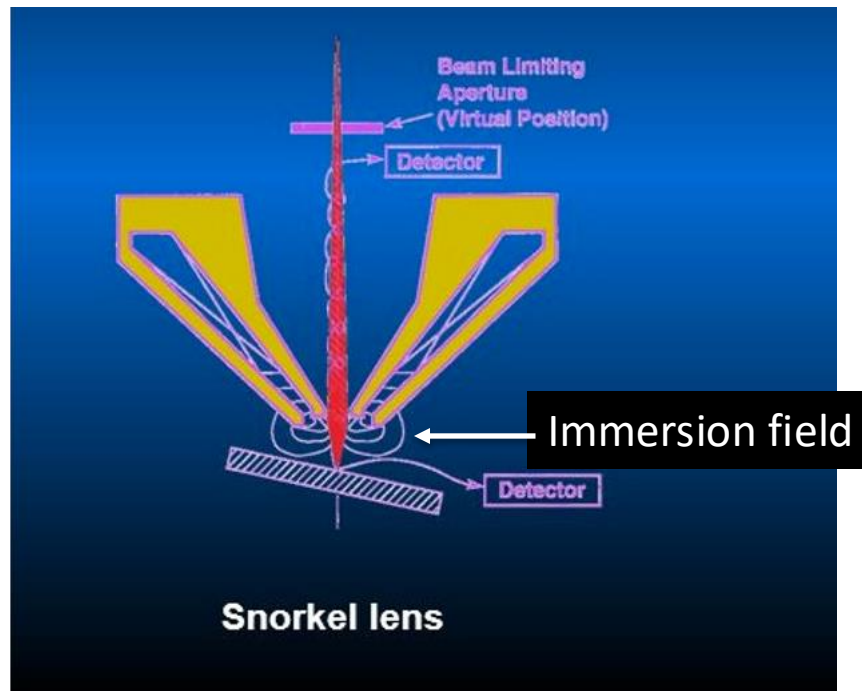
- a) The field increases the energy of the electron.
- b) The field increases the velocity of the electron.
- c) The field increases the momentum of the electron.
- d) The field does not change the electron energy.
- e) The field decelerates the electron.

Electromagnetic lens designs



Pinhole lens

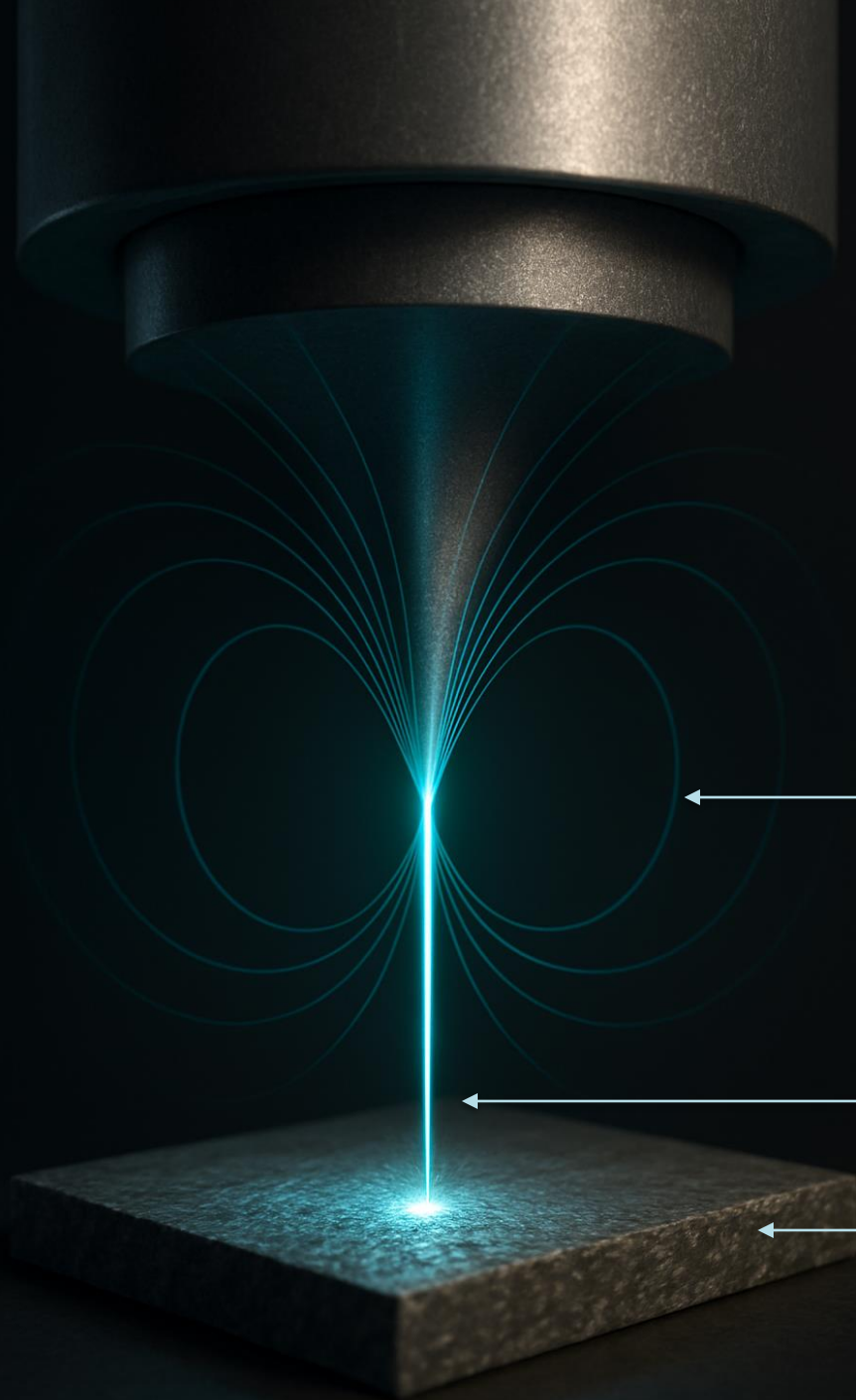
Magnetic field that focuses the beam is inside the pole piece.



Immersion lens

Magnetic field extends beyond the pole piece, and “immerses” the sample. This reduces the effective working distance, reduces aberrations and improves resolution.

Immersion lens



magnetic field extends out of the pole piece & immerses the sample

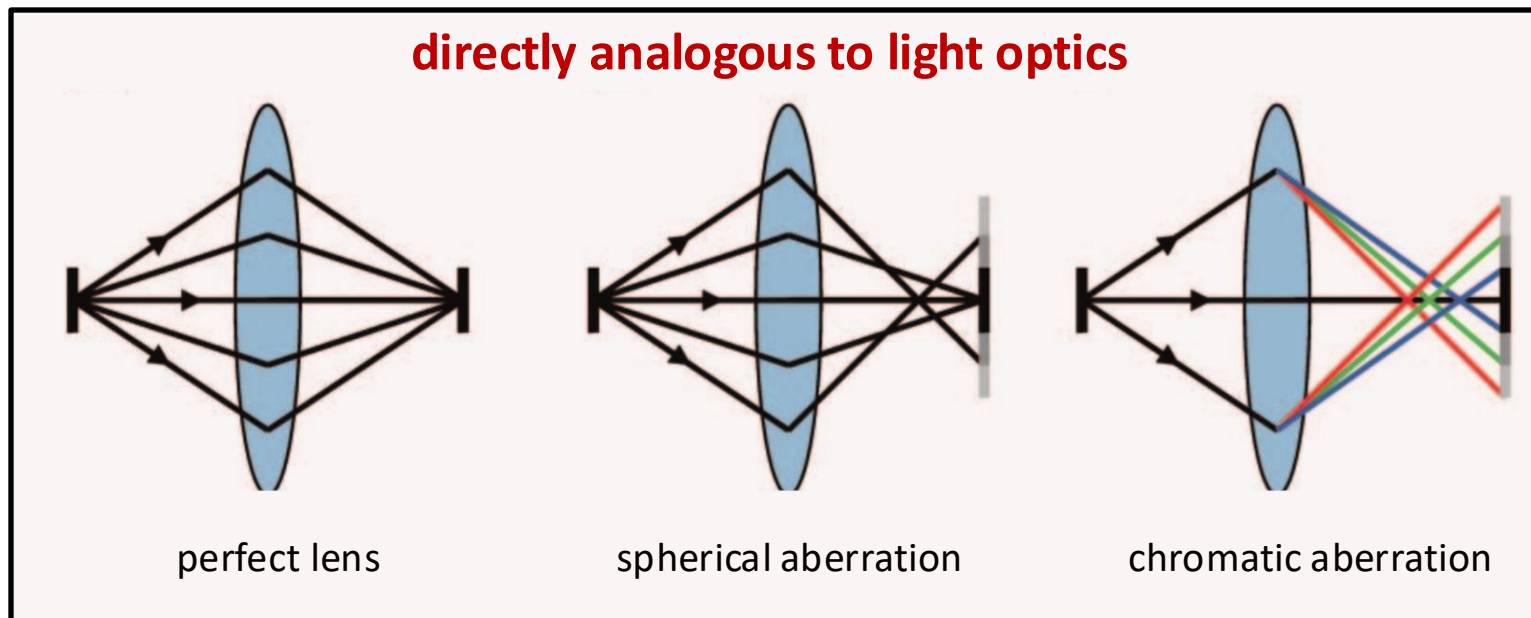
electron beam

sample

Electromagnetic lens: Aberrations

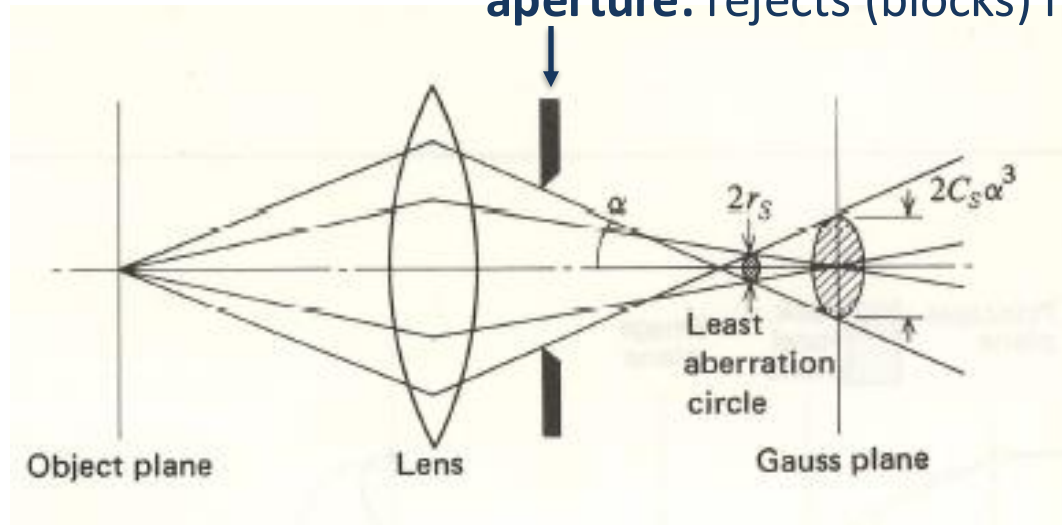
As with all lenses, electromagnetic lenses are subject to lens aberrations which increase the effective diameter of the probe - lowering spatial resolution

- **Spherical aberration:** electrons near the edge of the lens need to be focused more strongly than those near the centre
- **Chromatic:** due to thermal contributions, electrons emitted from the source are not all at the same energy (wavelength) so there is a spread of focus
- **Diffraction:** electrons being wave-like, small apertures can broaden the probe via diffraction effects
- **Astigmatism:** lack of radial symmetry of lens - non-circular probe



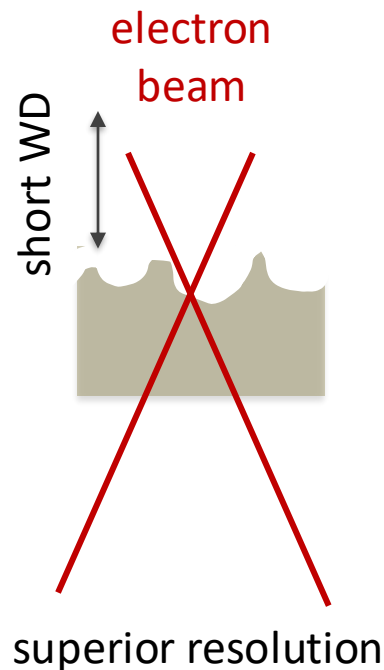
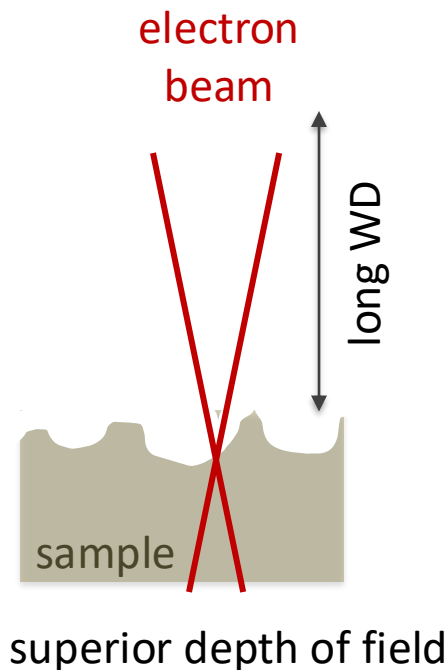
Spherical aberration, C_s

aperture: rejects (blocks) high angle electrons



C_s reduced by decreasing aperture diameter, which also limits the electron beam current.

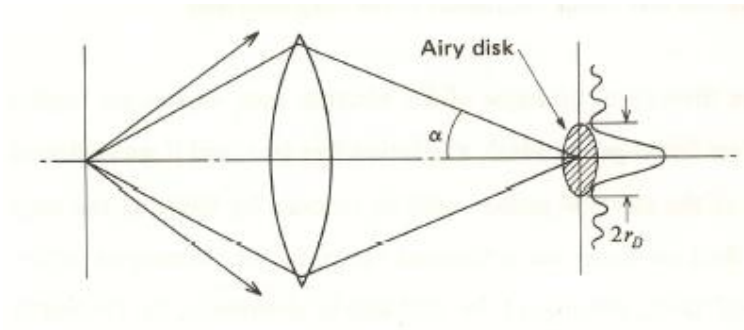
An SEM operator can use the aperture size to control resolution and electron beam current.



C_s is also reduced by increasing the magnetic field of the electromagnetic lens (ie: by increasing the beam convergence angle and operating the SEM at short “working distance”, WD).

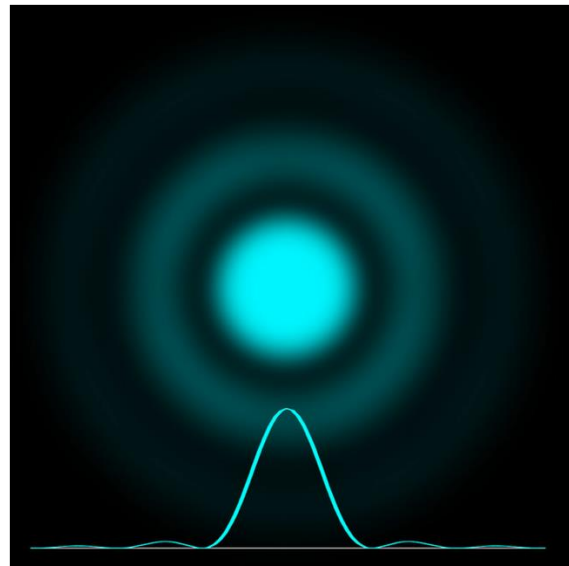
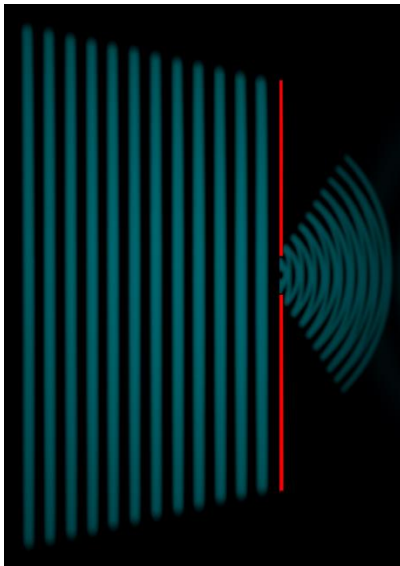
An SEM operator can use the working distance to trade off resolution for depth of field.

Diffraction aberration



As the aperture diameter decreases, electrons will diffract due to their wave-like properties

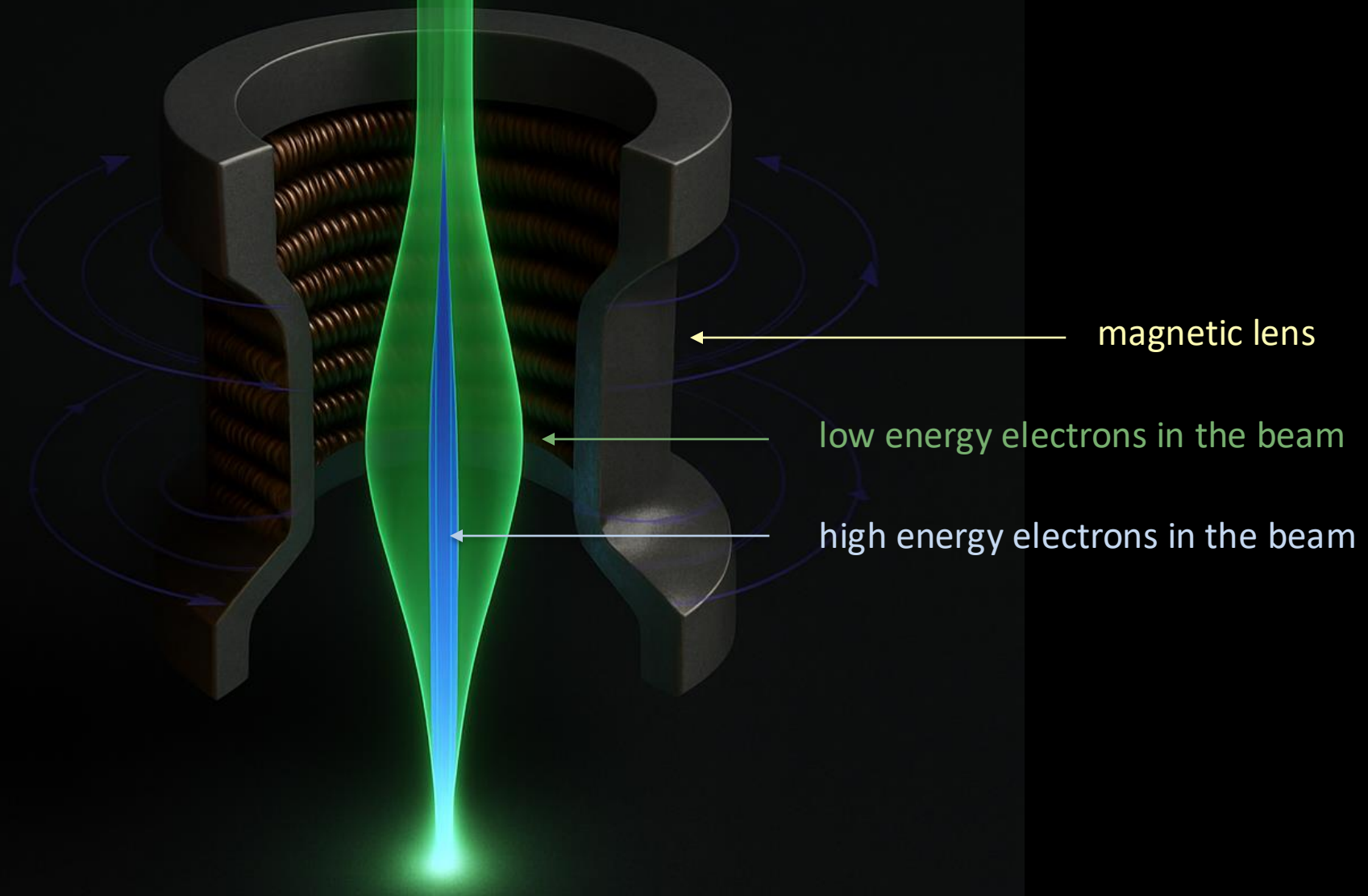
r_D reduced by increasing aperture diameter (i.e., if the aperture is too small, resolution is limited by diffraction)



there is an optimum aperture diameter (typically $\sim 10 \mu\text{m}$ in an SEM) that yields best resolution (ie: smallest electron beam diameter) by optimising both diffraction & spherical aberrations

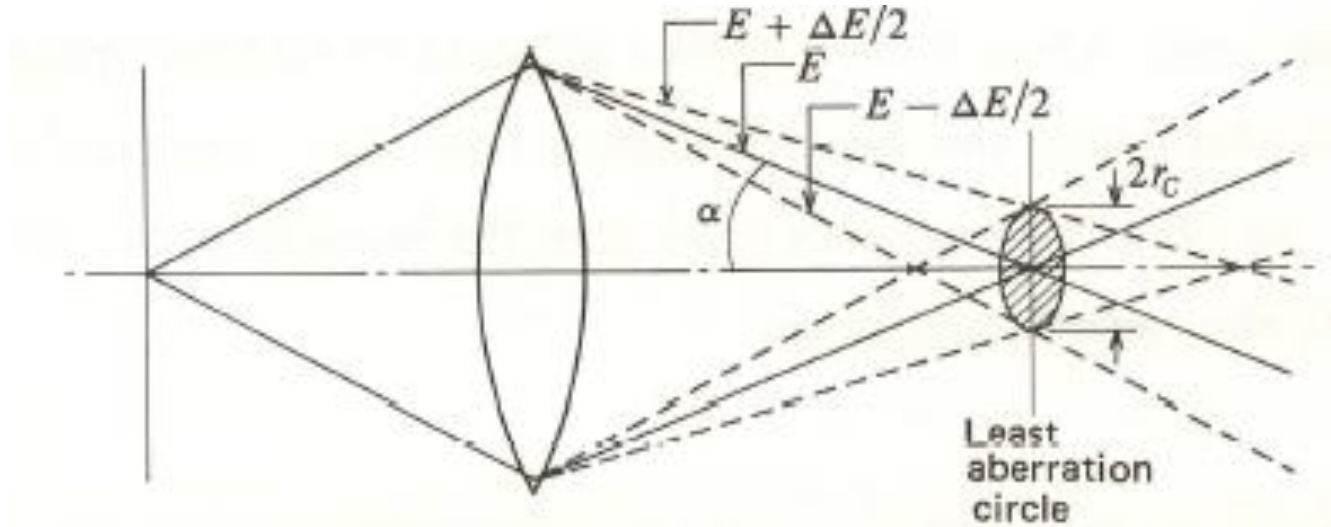
Reality check: the wavelength of electrons in an SEM is typically $< 1 \text{ nm}$. Despite the small wavelength, micron-sized apertures can still generate substantial diffraction aberrations due to beam scatter at the aperture. Factors such as the curvature of the aperture edge, surface roughness, and imperfections in the aperture geometry play a significant role in this diffraction. Additionally, the curvature of the electron wavefront as it interacts with the aperture also contributes to the overall diffraction and aberration effects.

Chromatic aberration



typical electron source
energy spread ~ 1 eV

Chromatic aberration, C_c

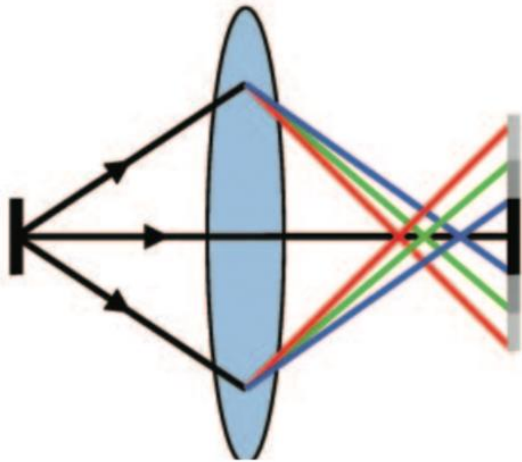


Lorentz force exerted on an electron by magnetic field **B**: $\mathbf{F} = q \mathbf{v} \times \mathbf{B}$

Electrons have a range of energies (i.e. “energy spectrum”) due to the temperature of the electron source (and also fluctuations in the accelerating voltage caused by noise in the power supply).

[energy spectrum] $\rightarrow$ [velocity spectrum] $\rightarrow$ [range of forces **F**] $\rightarrow$ [range of focal lengths f]

Chromatic aberration



Is this analogy to light optics correct?
i.e. focal length of [red beam] < [green] < [blue] ?

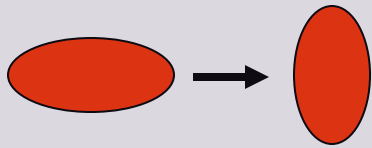
Lorentz force exerted on an electron by magnetic field **B**: $\mathbf{F} = q \mathbf{v} \times \mathbf{B}$

Answer: Yes, due to inertia. i.e. Even though the Lorentz force is stronger for the faster (higher-energy) electron, its inertia (momentum) scales faster with energy than the sideways Lorentz force does. So, the net curvature decreases with increasing electron energy.

Correct intuition: Higher energy = higher forward momentum → more “resistant” to bending.

Astigmatism aberration

Astigmatism Present

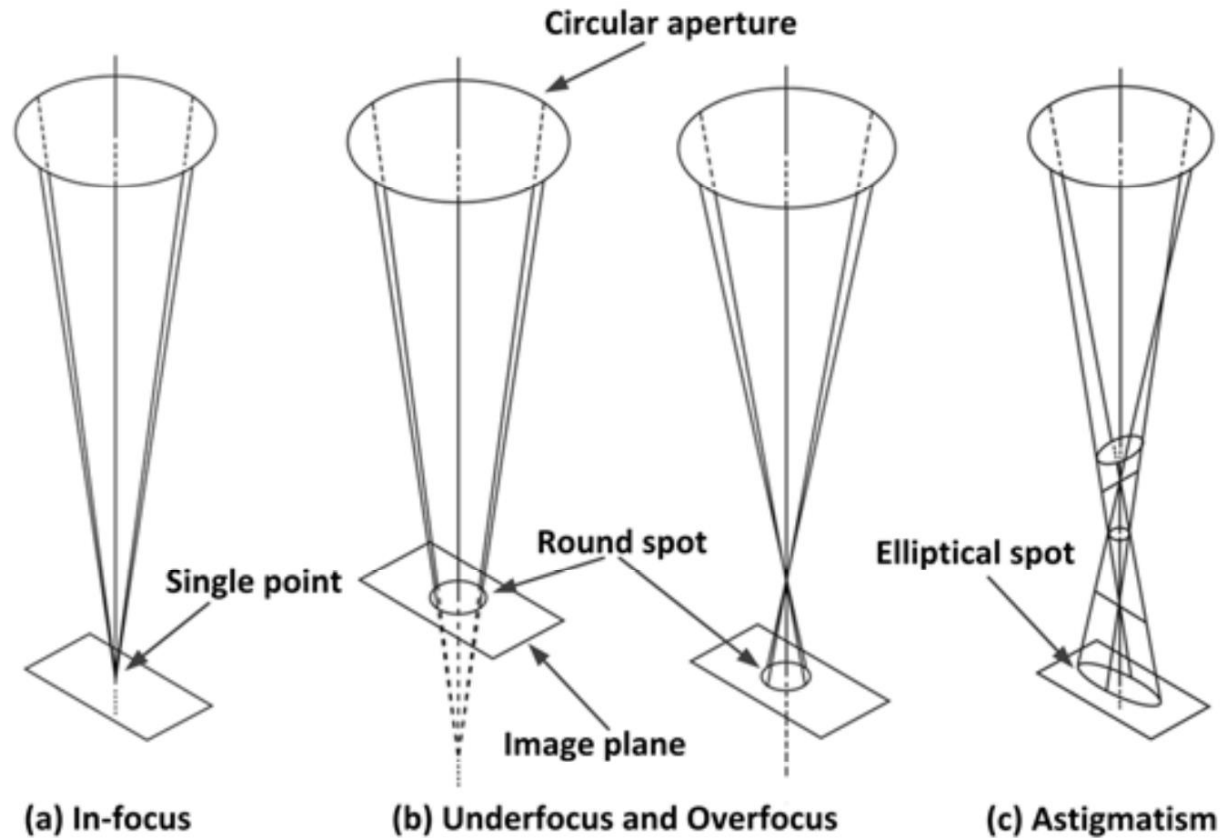


rocking focus

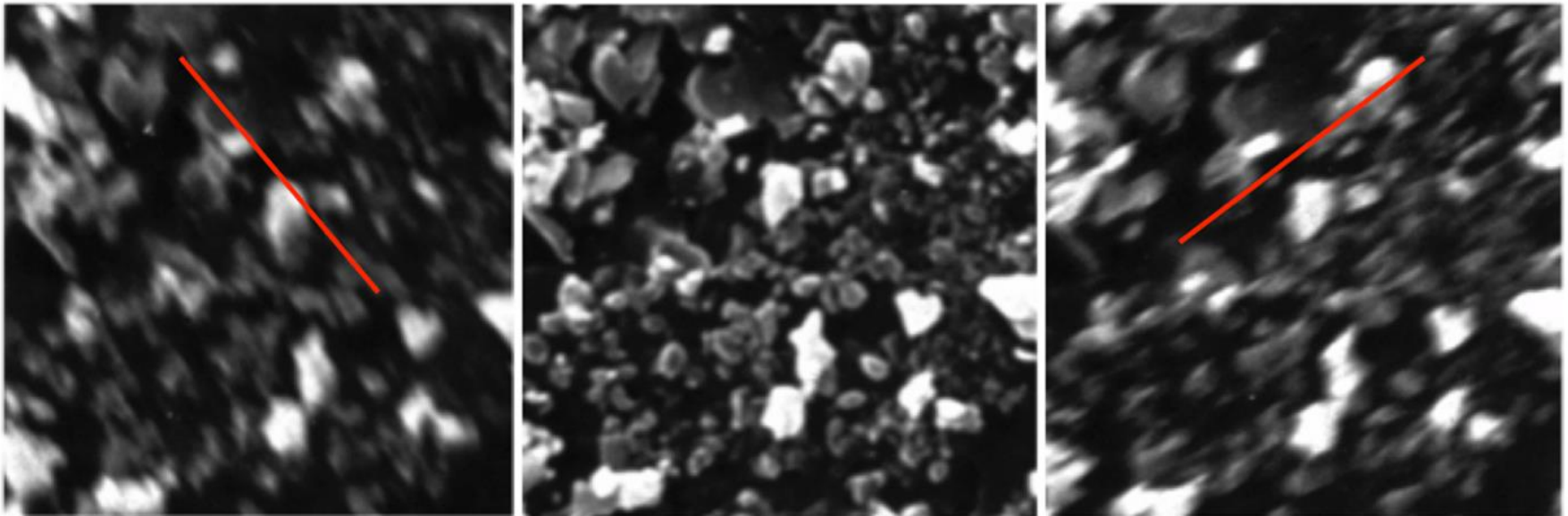
Astigmatism Corrected



rocking focus

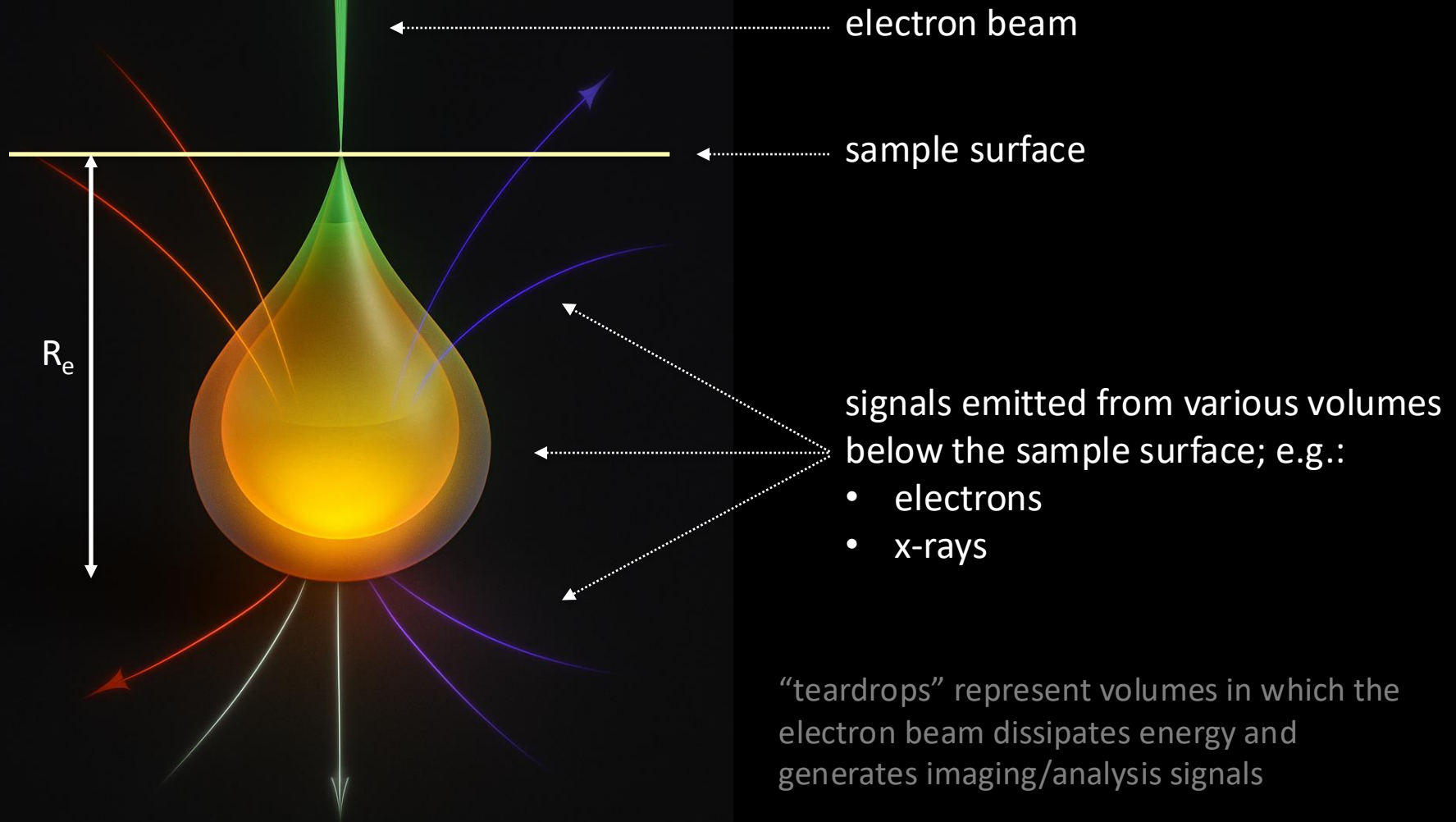


Astigmatism aberration



- caused by lack of radial symmetry in the lens
- produces an elliptical electron beam spot rather than circular
- identified by 90° streaking as image rocked through focus
- corrected by stigmators that re-shape beam to a circular spot – either r , θ or x , y control to adjust orientation and strength of the correction field

Electron-solid interactions. Imaging/analysis signals.



“teardrops” represent volumes in which the electron beam dissipates energy and generates imaging/analysis signals

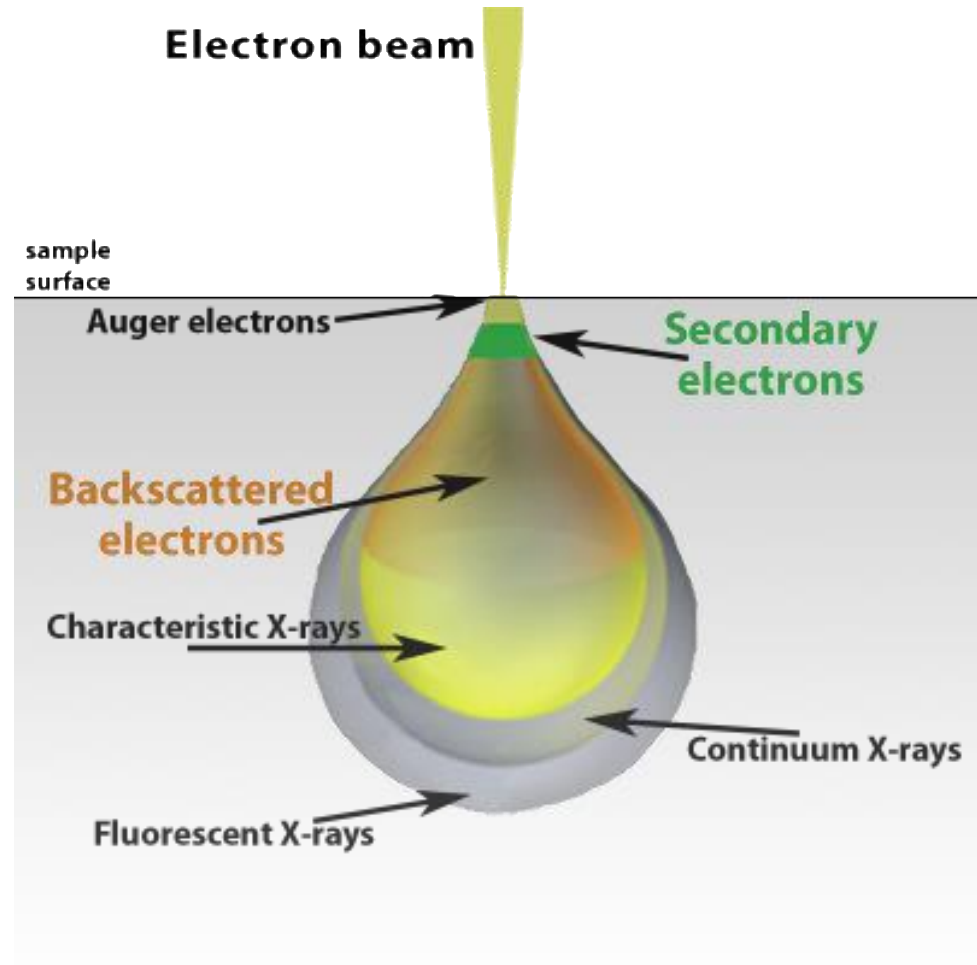
R_e = electron range ~ 10 nm to ~ 1 μ m, based on electron beam energy

Interaction of the electron beam/probe and the sample

this is covered in this week's tutorial

- the total energy of the incident electrons must be dissipated (in a bulk sample) and energy must be conserved
- energy loss processes generate emission signals that can be used to form an image as the electron beam scans across the specimen
- each of these signals carries different information about the sample

Each emissive signal originates from a different volume (and different maximum depth) within the sample

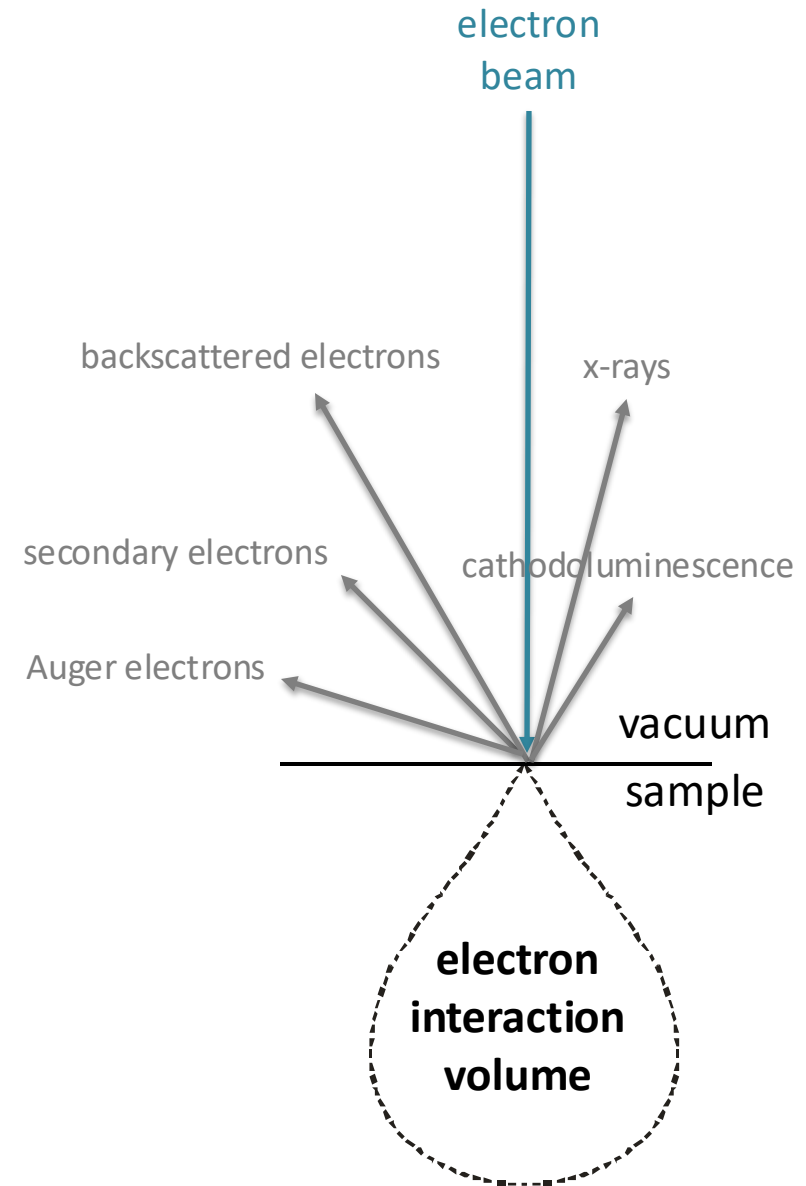


Main energy loss processes in solids

specimen heating: *generation of phonons*

breaking of chemical bonds that bind atoms:
(1 -10 eV) produces secondary electrons and cathodoluminescence

ionization of core shell electrons:
(keV): produces X-rays



Electron Scattering Mechanisms

Elastic Scattering (due to interactions with nuclei)

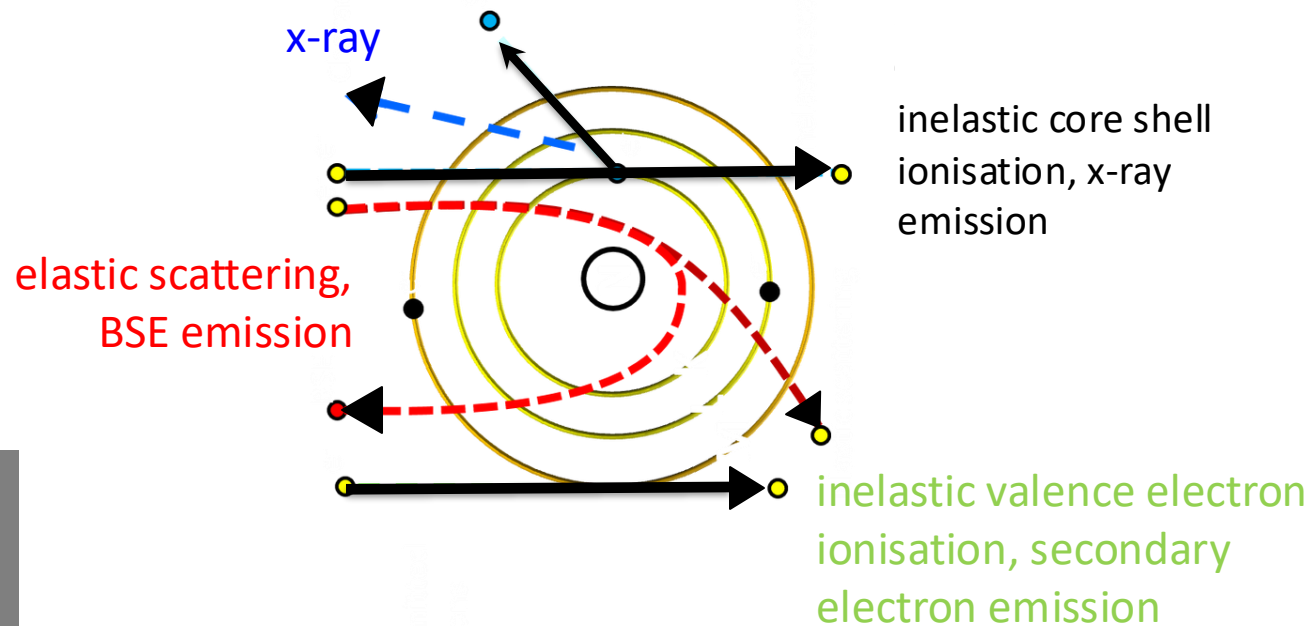
Change electron trajectories with virtually no loss in energy

Causes: (i) electron beam spread (“straggle”) in the sample, &
(ii) backscattering of electrons

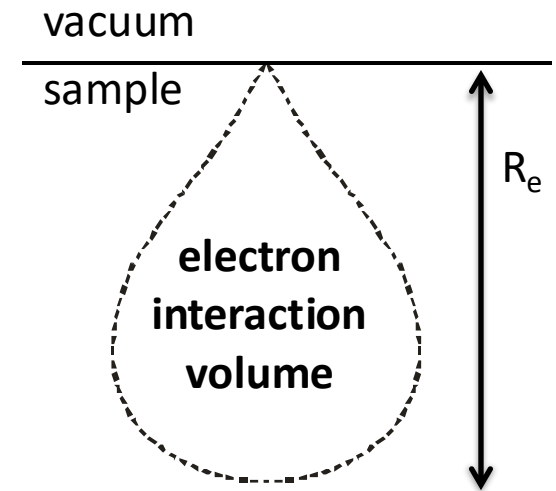
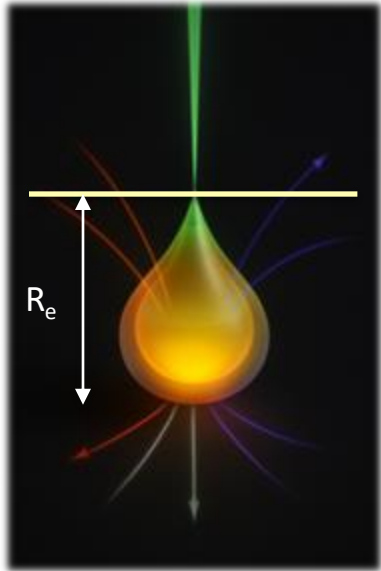
Inelastic Scattering (due to electron-electron scattering)

Causes significant loss of energy with no change in trajectory

Generation of heat, secondary electrons, cathodoluminescence and X-rays



Primary (beam) electron range (R_e)



atomic weight [g/mol]

energy of beam [keV]

$$R_e(nm) = \frac{27.6 A E_o^{5/3}}{Z^{8/9} \rho}$$

atomic number

density g/cm³

Electron emission:

Backscattered Electron (BSE) imaging in Scanning Electron Microscopy (SEM)

Papers on Canvas (useful for this week's tute)

\Canvas\Modules\Tutorials...\Tute 2...\07. CASINO - intro.pdf

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Original Papers

CASINO : A New Monte Carlo Code in C Language for Electron Beam Interaction
—Part I: Description of the Program

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Summary: This paper is a guide to the ANSI standard C code of CASINO program which is a single scattering Monte Carlo simulation of electron trajectory in a solid specially designed for low-beam interaction in a bulk and thin foil. CASINO can be used either on a DOS-based PC or on a UNIX-based workstation. This program uses tabulated Mott elastic cross sections and experimentally determined stopping powers. Function pointers are used for the most essential routine so that different physical models can easily be implemented. CASINO can be used to generate all of the recorded signals (x-rays, secondary, and backscattered) in a scanning electron microscope either as a point analysis, as a linescan, or as an image format, for all the accelerated voltages (0.1–30 kV). As an example of application, it was found that a 20 nm Guinier-Preston Mg₂Si in a light aluminum matrix can, theoretically, be imaged with a microchannel backscattered detector at 5 keV with a beam spot size of 5 nm.

Key words: Monte Carlo simulation, low-energy scanning electron microscope, backscattered electrons, image simulation, line profiles, x-rays $\phi(\rho z)$ computation, Mott elastic cross section, experimental stopping power

1. Introduction

Nowadays, new compounds and alloys are designed to answer specific needs and specific applications. In general, the new high-strength alloys contain a very fine scale microstructure, less than micrometer scale grain size, and precipitates < 0.05 μm . The development of such new materials must be followed by both crystallographic and chemical characterization to yield a full understanding of the microstructure and property relationships. In this perspective, electron microscopy (scanning and transmission) is the most complete and powerful instrument that could be used. Scanning electron microscopy (SEM) has the advantage that sample preparation is generally more straightforward compared with the preparation of the electron transparent region needed for the transmission electron microscope (TEM).

Combining a field emission source into an SEM (FESEM) results in a high current density into a very small probe (1 nm @ 30 keV). The FESEM can be used efficiently to characterize the new high-strength alloys as well as complex semiconductor multilayers. In addition, the FESEM opens new opportunities for imaging and microanalysis at low voltage (< 5 kV) such as the examination of non- or poorly conducting materials (Joy and Joy 1996), the imaging dopant concentration in semiconductor (Perovic 1995), quantitative x-ray analysis with resolution < 100 nm, and the imaging of phases < 20 nm (Gauvin *et al.* 1995a). Traditionally, Monte Carlo programs are used to exploit and understand fully the capabilities of electron microscopes (Newbury and Yakowitz 1975) at energy > 10 keV. However, for low energy ($E < 5$ keV), because the first Born approximation is no longer valid, a new generation of Monte Carlo programs is urgently needed. A careful investigation of electron-solid interaction models must be developed before any conclusion can be given.

In this paper, we fully describe a complex Monte Carlo program specifically designed for low-beam interaction that can be used to generate all of the recorded signals (x-rays, secondary and backscattered) in an SEM. This program can also be efficiently used for the range of acceleration voltage found on an FESEM (0.1–30 kV). The goal of this paper is to provide users with a complete description of the most important procedures and features of this Monte Carlo program. Description of the available physics will also be presented. In addition, a full description of the most important routines will be given. This section is more programmer-oriented. As an example of application, the possibility of imaging < 100 nm Mg₂Si precipitates in an aluminum matrix (Al6061 alloy) will be presented. The sources code and the executable file is now available on the Internet at the address <http://casino.gme.usherbrooke.ca/casino>

Address for reprints

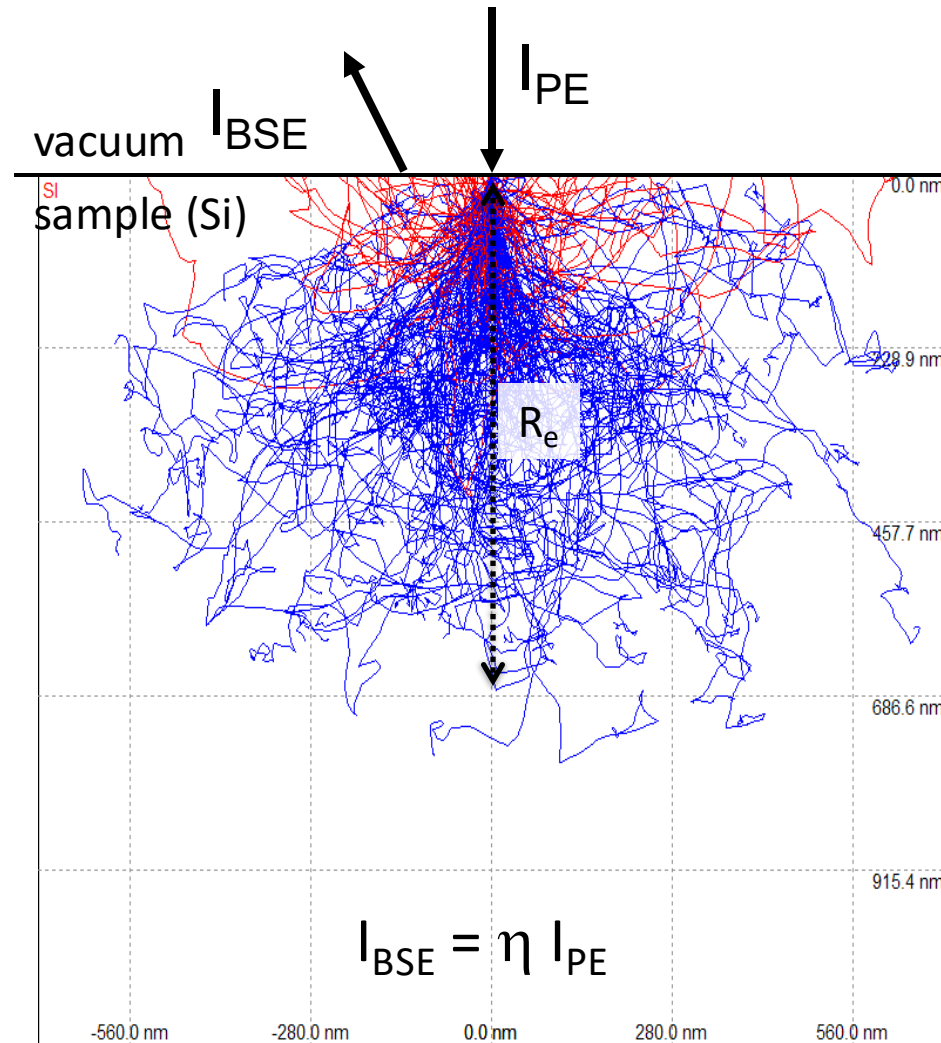
Pierre Hovington
Département de Génie Mécanique
Université de Sherbrooke
2500 Boulevard Université
Sherbrooke, Québec
Canada, J1K 2R1

2. Features of the Program

This Monte Carlo program contains over 50,000 lines of code in ANSI standard C language distributed over 20 files. It

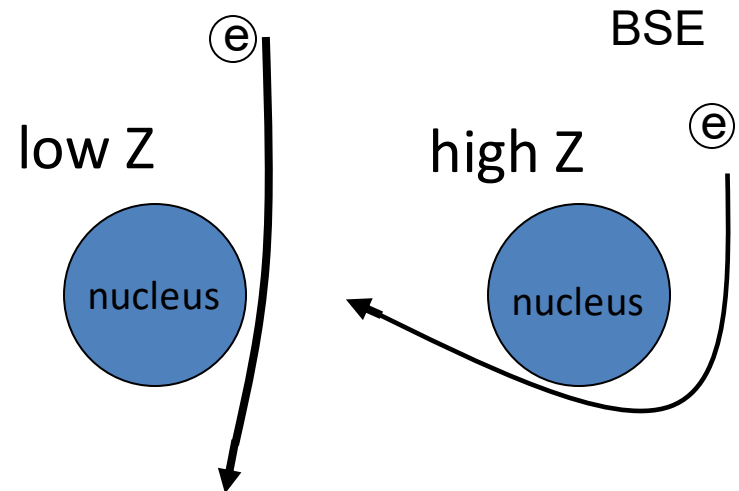
Backscattered Electrons: compositional imaging

this is covered in this week's tutorial



η is the BSE yield

Red trajectories are primary electrons that have been back-scattered out of the specimen.



larger Z increases the electron scattering efficiency, so η increases with mean Z

BSE compositional Imaging

$$I_{\text{BSE}} = \eta I_{\text{PE}}$$

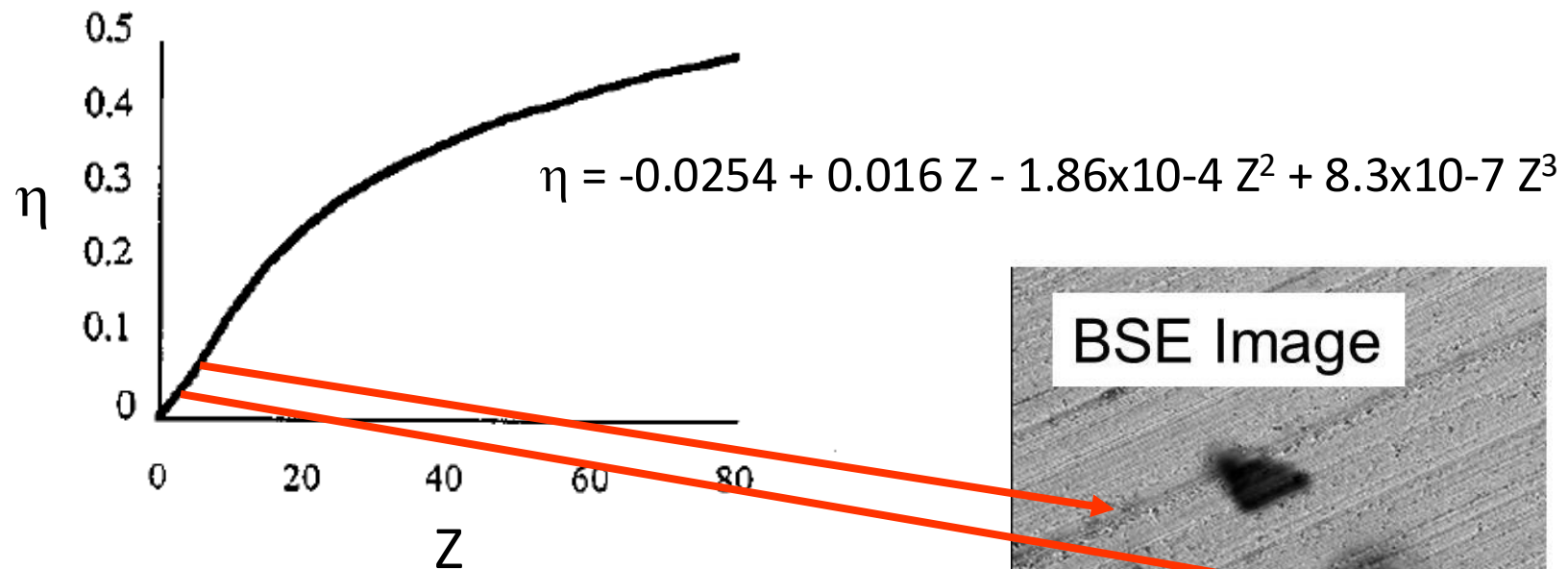
I_{PE} = primary electron beam current

I_{BSE} = backscattered electron (BSE) current

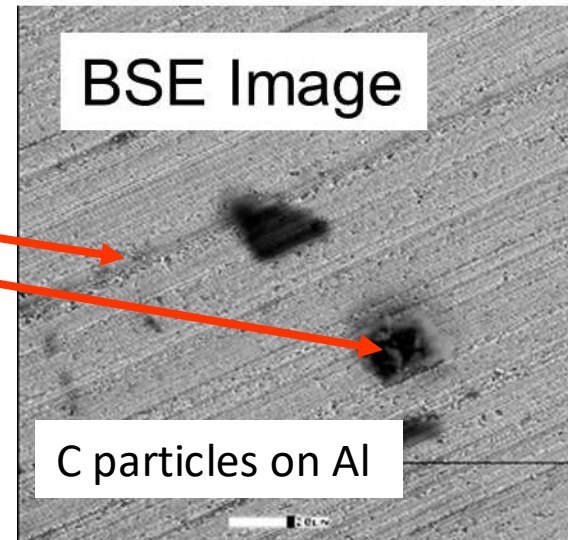
η = BSE coefficient (a.k.a. "BSE yield")

$$\eta = I_{\text{BSE}} / I_{\text{PE}}$$

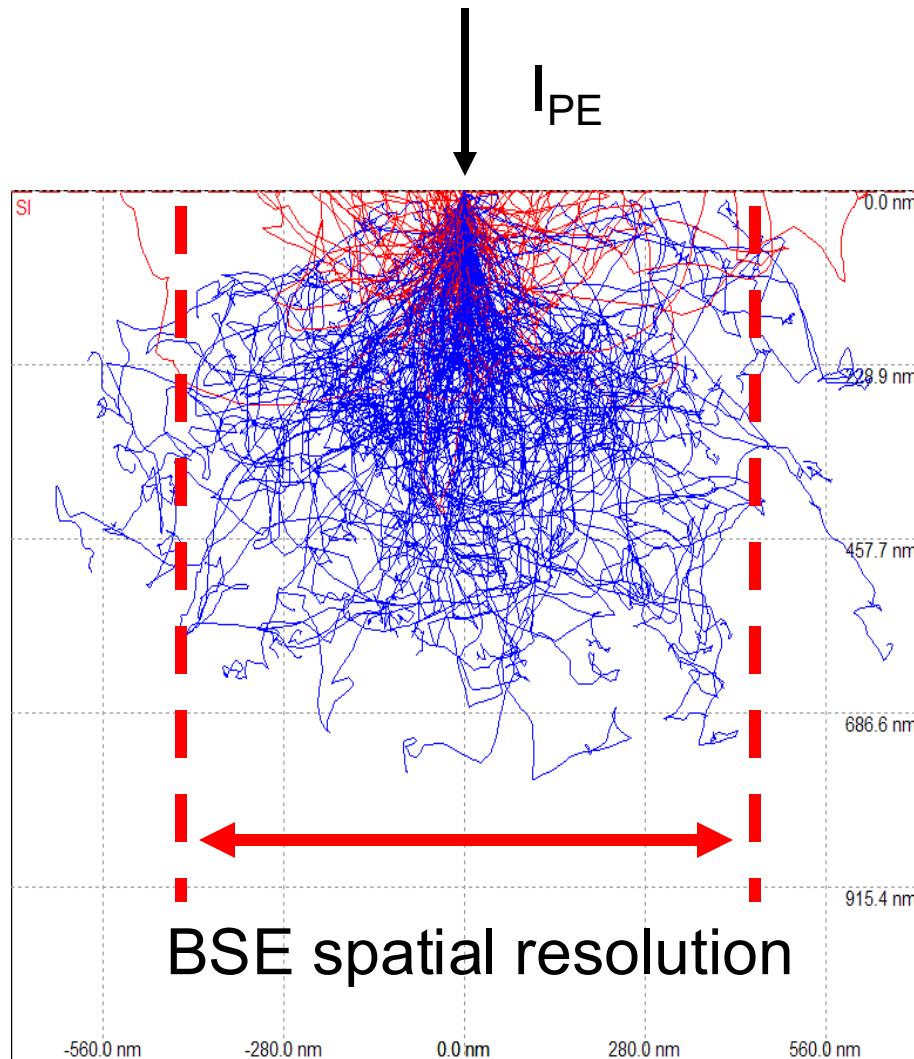
η increases with increasing atomic number (Z)



Dark carbon particles in BSE image have lower average Z than Al substrate



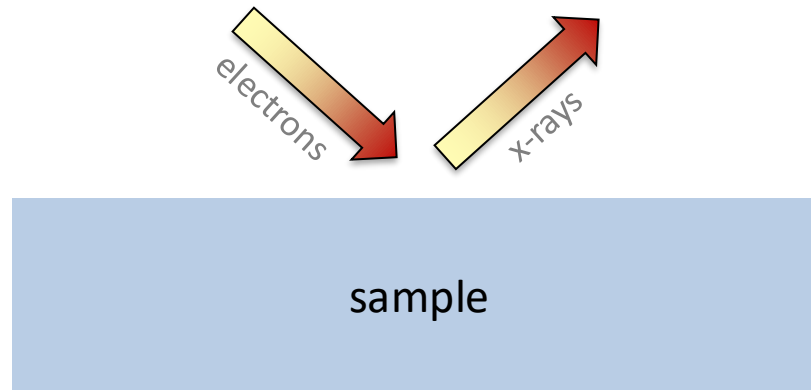
BSE Compositional Imaging: Resolution



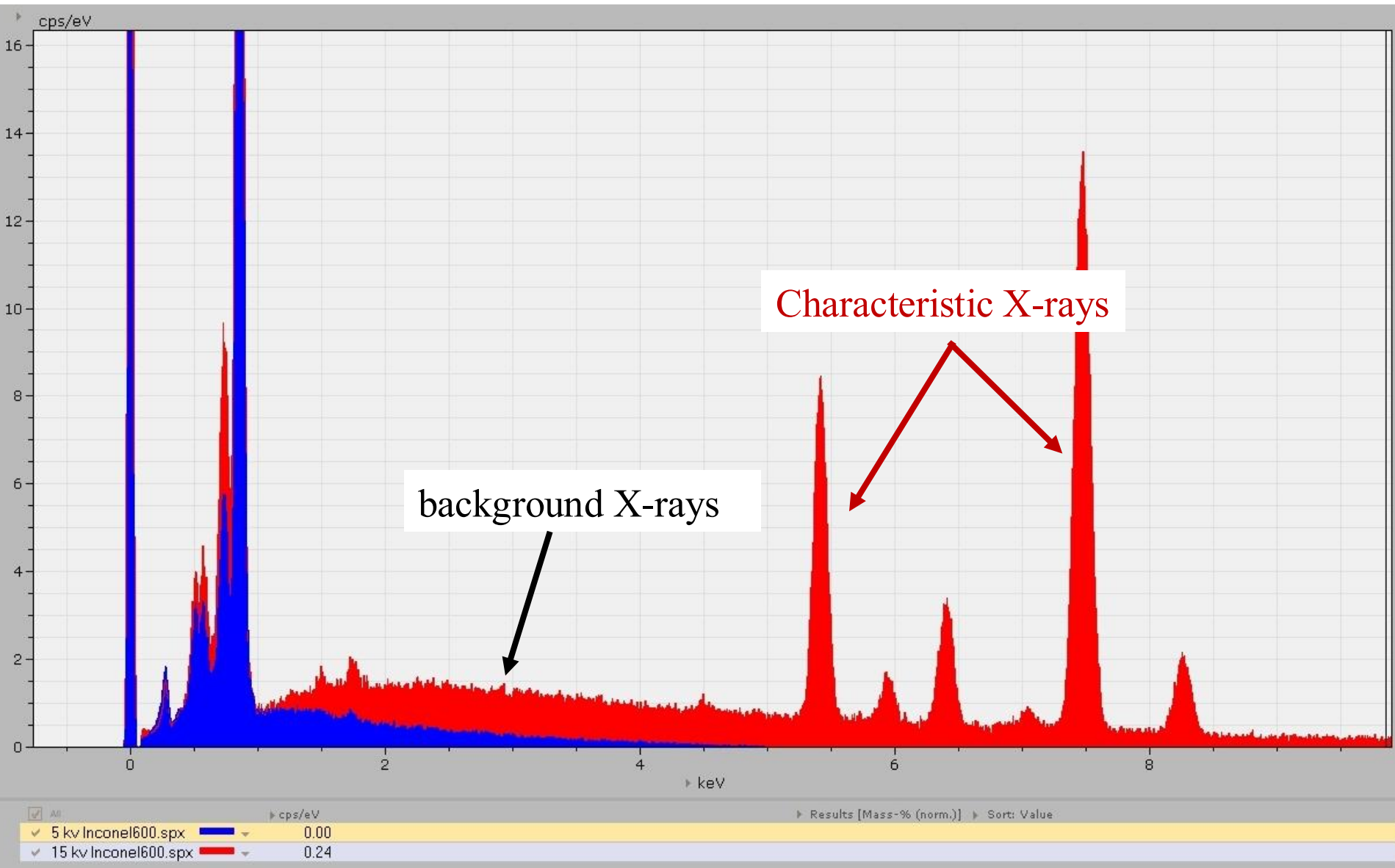
BSE image resolution is approximated by the lateral dimensions of the interaction volume rather than the electron probe diameter

Brief introduction to:

X-ray analysis in a scanning electron microscope;
Energy-dispersive x-ray spectroscopy (EDX)

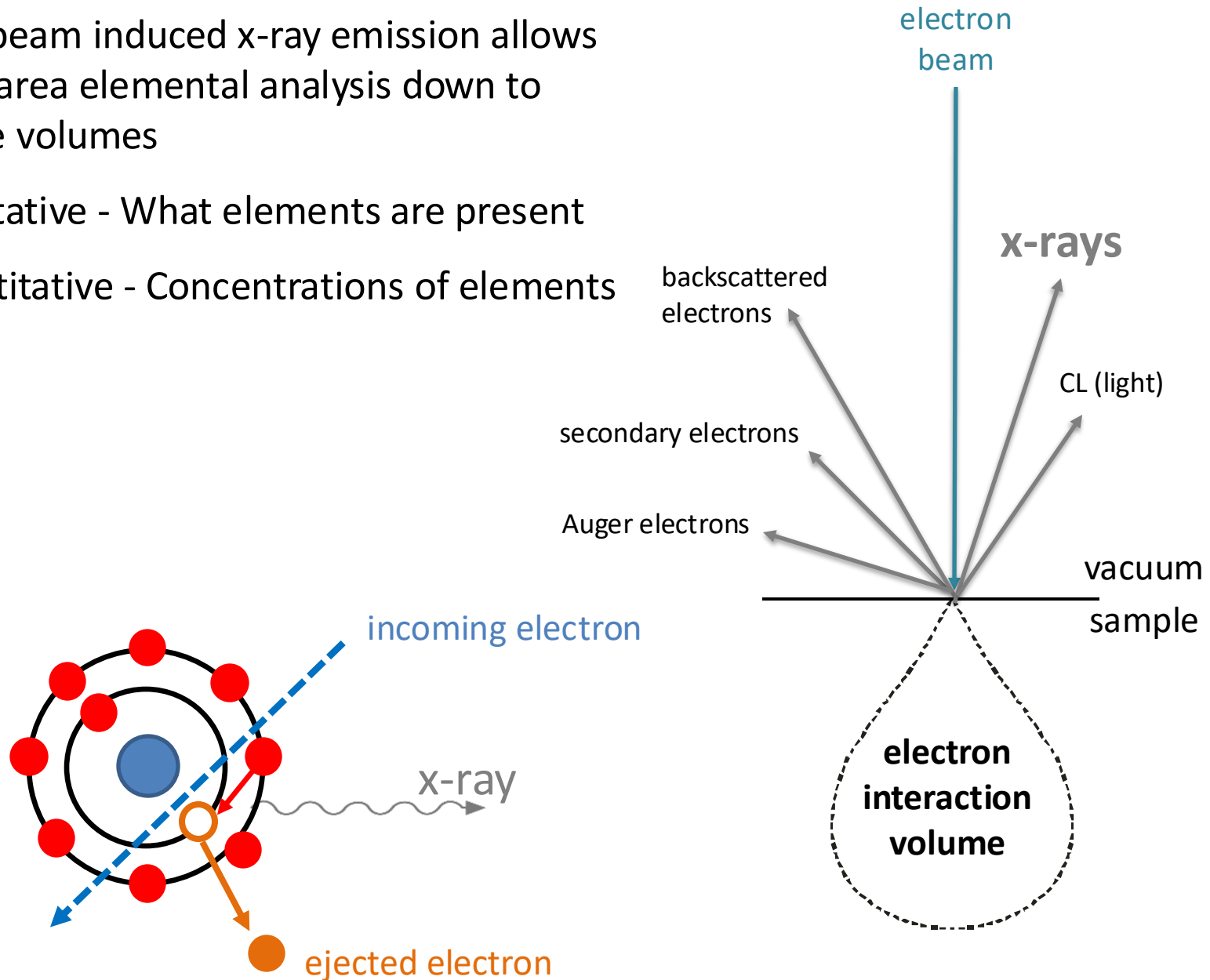


EDX of steel: 5keV (Blue) & 15 keV (red)

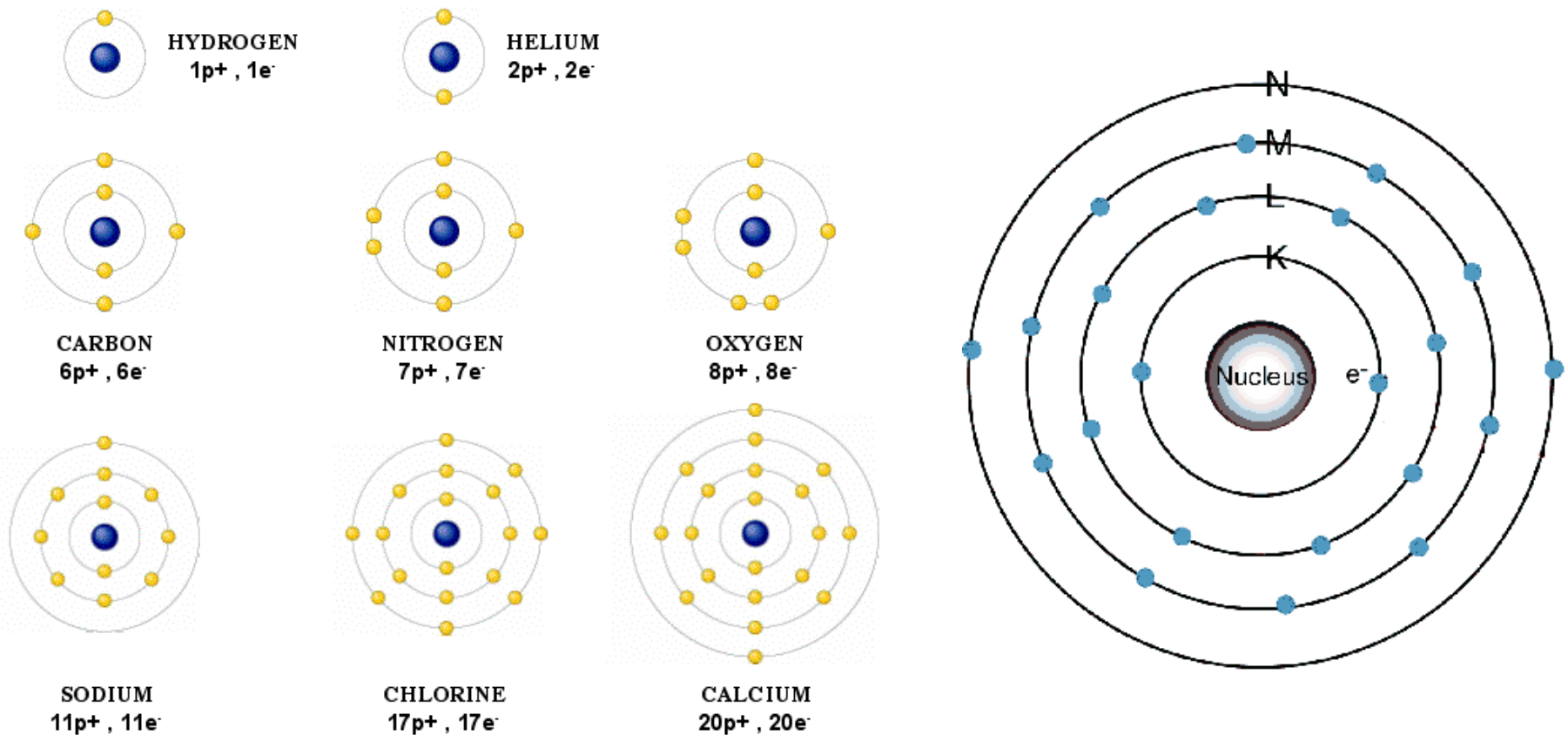


Characteristic x-rays: Microanalysis

- electron beam induced x-ray emission allows selective area elemental analysis down to nanoscale volumes
 - Qualitative - What elements are present
 - Quantitative - Concentrations of elements



Atomic structure

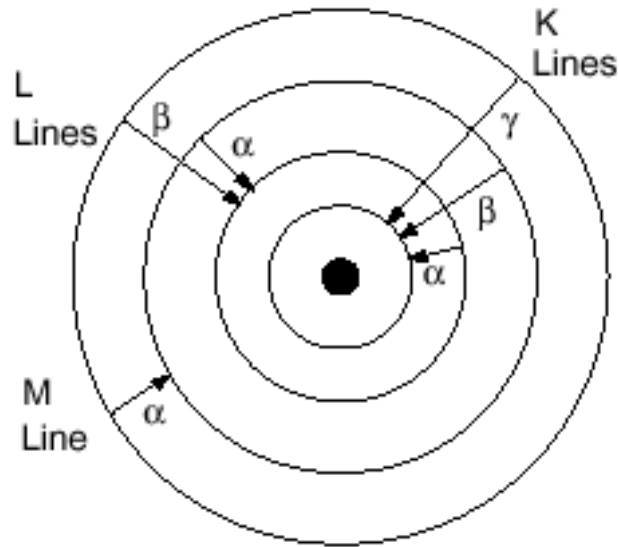


X-ray lines are labelled K, L, M... These letters denote the orbital that is ionised (ie: the orbital that an electron transitions to upon x-ray generation)

Energy required to remove the electron from an orbital is known as the “ionisation potential” or “ionisation energy”, generally around 0.1 to 50 keV

Characteristic x-rays

- energy of the X-ray is determined by the energy difference between the atomic levels involved in the electronic transition
- each element has a characteristic set of X-rays and X-ray spectra can be used to identify elements from nano-scale volumes



X-ray notation

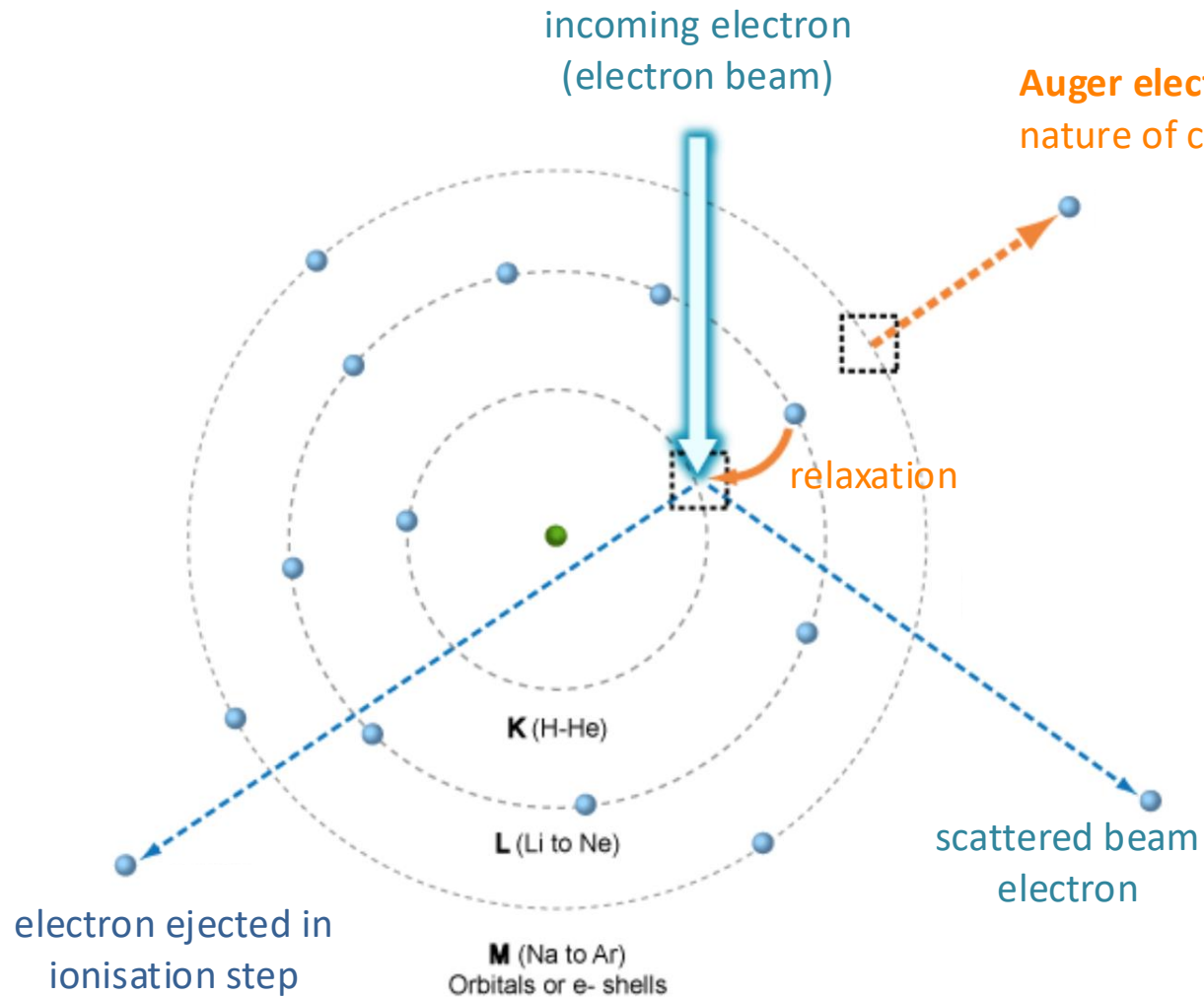
K x-rays are transitions that terminate on **1s level**

L x-rays are transitions that terminate on **2s, 2p levels**

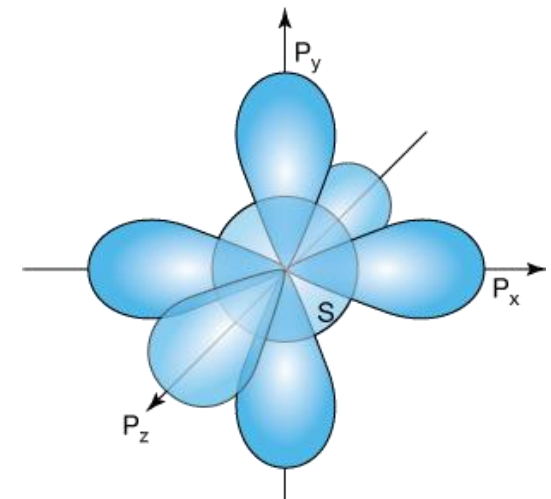
M x-rays are transitions that terminate on **3s, 3p, 3d level**

Greek symbols denote the initial level of the electron. Example of full notation:
Carbon K_{α} x-ray, or K_{α} “emission line”

Auger electron emission



The Auger electron emission process is triggered by internal electron-electron interactions within an atom – these interactions pertain to overlaps between the electron wavefunctions within the solid



electron orbitals are representations of the wavefunctions of electrons bound to a nucleus

Imaging and analysis signals

Backscattered electrons

Mean atomic number

X-rays

Elemental composition

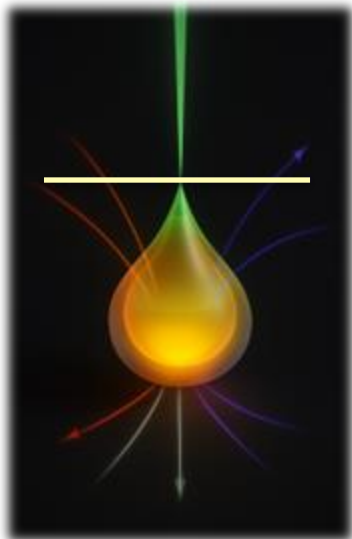
Limited info on chemical bonds

Auger electrons

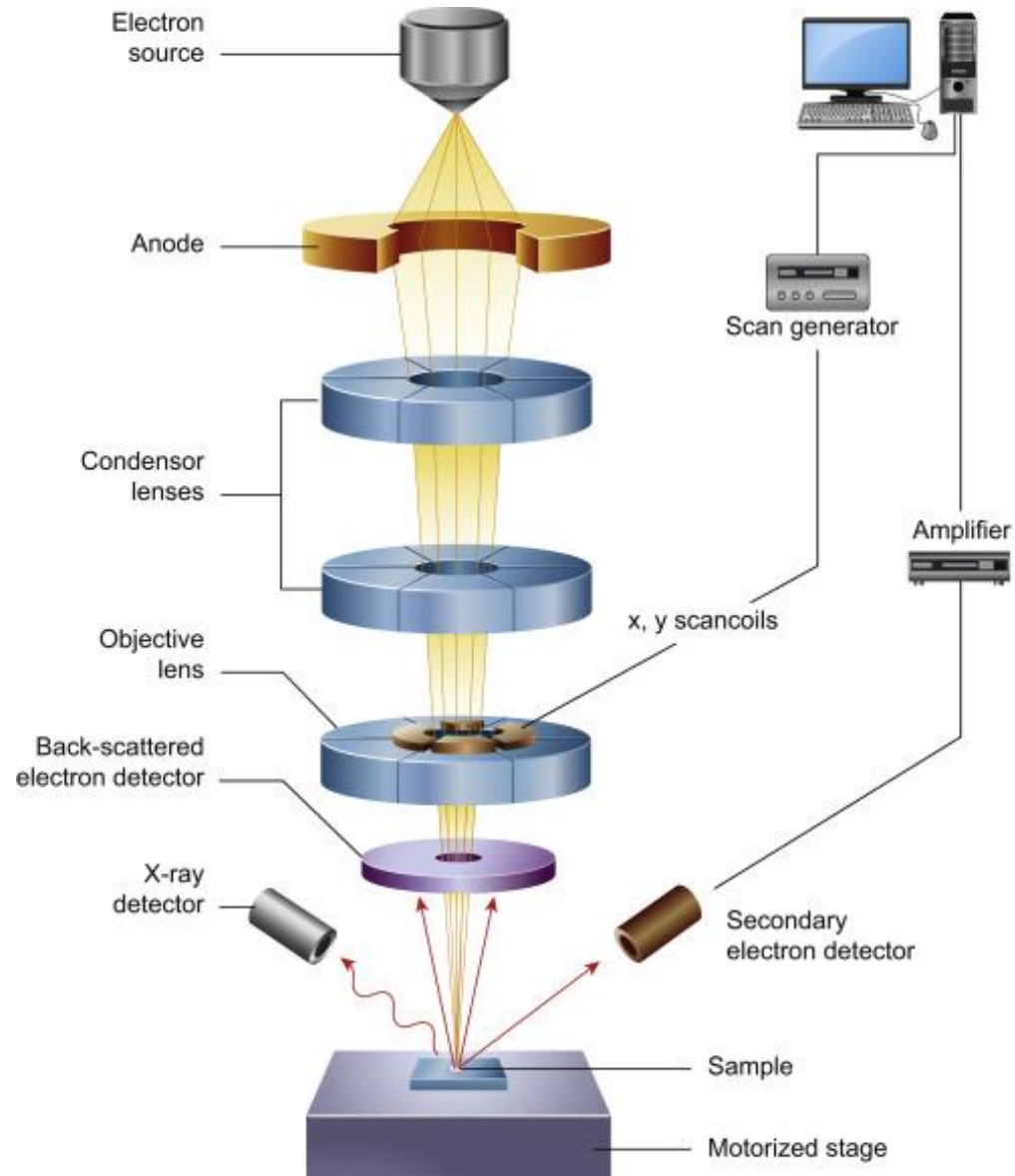
Elemental composition

More info on chemical bonds

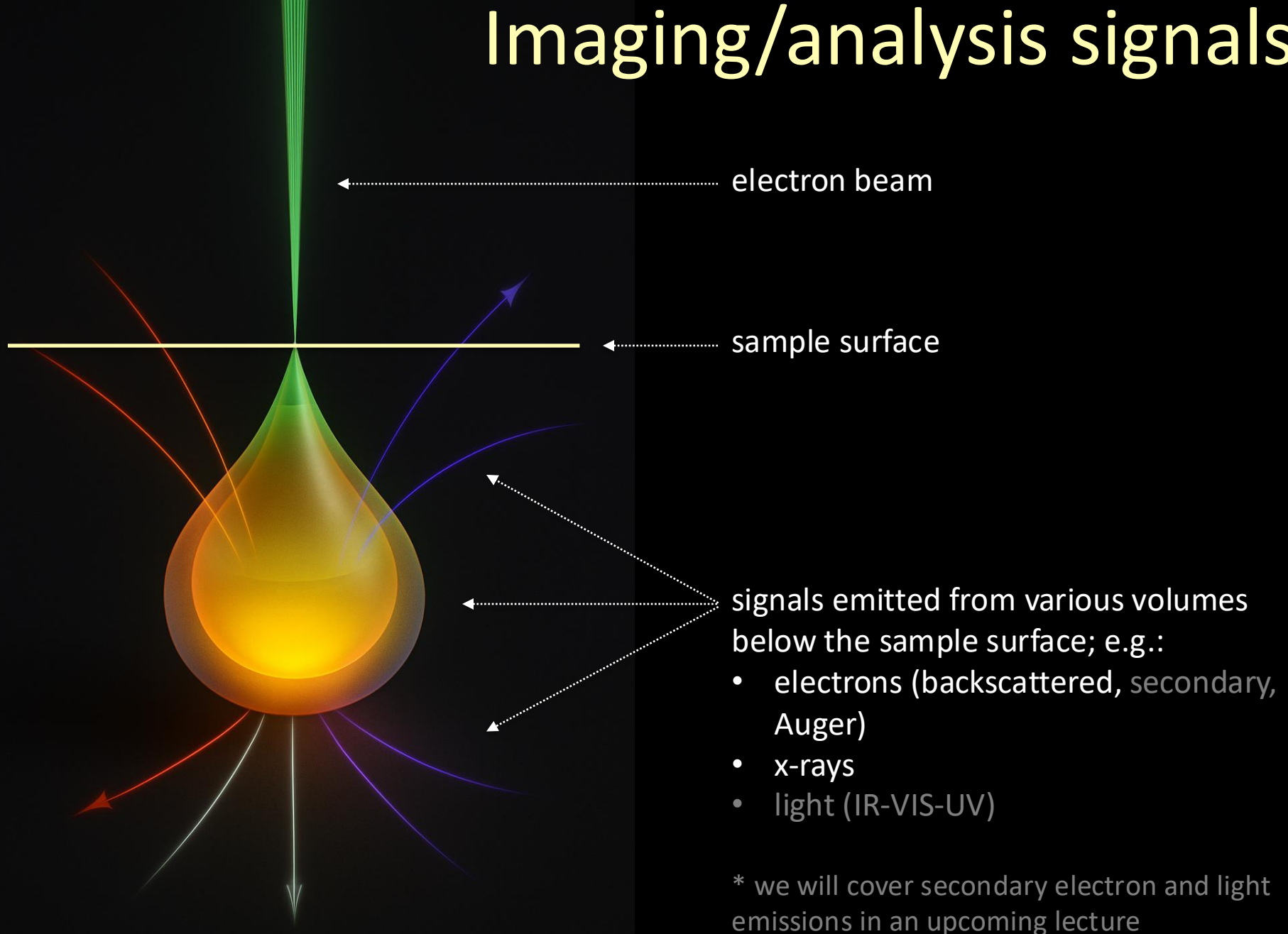
High surface sensitivity



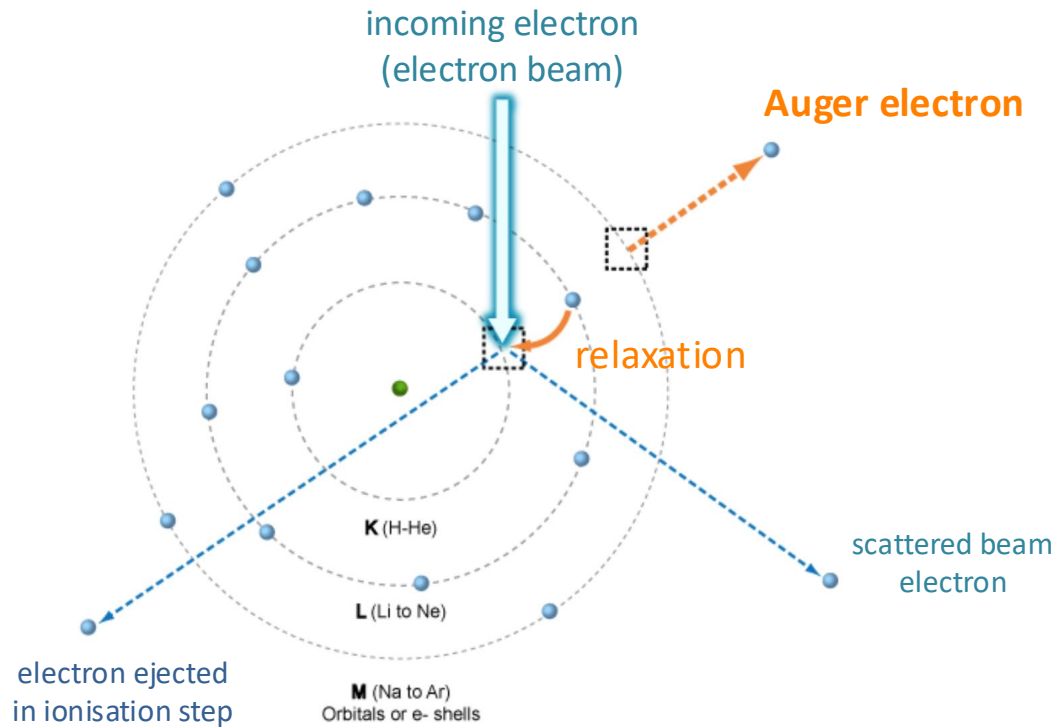
homework: why are Auger electrons more "surface-sensitive" than x-rays emitted from a sample in SEM?



Imaging/analysis signals



Sample class test question



“Auger electrons provide more information on chemical bonding than x-rays emitted from a sample in an electron microscope” – Is this statement true? Why?

- a) No, because inner shell electrons participate in chemical bonds rather than outer shell electrons.
- b) No, because outer shell electrons participate in chemical bonds rather than inner shell electrons.
- c) Yes, because Auger electrons conserve energy by forming chemical bonds.
- d) Yes, because Auger electrons are entangled with nuclei in the sample.
- e) Yes, because the energy of outer shell electrons is modified by chemical bonds.
- f) Yes, because the energy of outer shell electrons is modified by neighbouring atoms.